

Tax Reporting Simplification and Firm Behavior *

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Abstract: Simplifying regulation and reducing bureaucratic costs are widespread policy objectives, yet evidence on their economic effects and on firms' behavioral responses remains limited. We study whether and how a major regulatory simplification in the tax domain affects real business activity. Our setting leverages the European Union's 2021 VAT reform, which substantially simplified VAT reporting and remittance for firms exporting to EU member states. Combining novel transaction-level administrative data covering more than 150 million observations with a difference-in-differences design, we find that the simplified reporting significantly increased real economic activity. Treated firms execute roughly 34% more transactions, diversify their product scope, and expand into additional countries. Survey evidence confirms that firms perceive the reform as reducing compliance costs, and our mechanism tests are consistent with lower prices, reduced market-entry costs, and freed-up firm resources. However, we also document an unintended reporting consequence of the reform. For non-EU exporters, the simplified reporting applies only to transactions valued below EUR 150. After the reform, the distribution of transactions exhibits pronounced excess mass just below this threshold and a corresponding hole above it. Average prices and parcel weights per transaction decline after the reform, consistent with a strategic reporting response in which firms misreport values and split transactions into multiple parcels to qualify for the simplified regime.

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1 Introduction

Simplifying regulation and reducing bureaucratic costs for businesses are stated goals of economic policy in countries worldwide.¹ Policymakers seek to reduce regulatory frictions and administrative burdens, arguing that simpler, less bureaucratic rules lower compliance costs and ultimately stimulate the economy. One area where calls for simplification are particularly prevalent is business taxation (European Commission 2020; OECD 2025a; Baer and De Mooij 2026). The tax environment in which firms operate is highly complex, and prior work finds that this complexity affects firm behavior along several margins (Zwick 2021; Amberger et al. 2025). Yet, despite the widespread policy objective of simplifying the tax system, empirical evidence on whether and how firms respond when complexity is actually reduced is limited. In this paper, we address this gap in the literature by studying whether simplified tax regulation affects business activity. Specifically, we study how a European Union (EU) reform that considerably simplified cross-border value-added tax (VAT) payments affects firms' international transactions, including transaction frequency, sales volume, and the number of traded products and destination countries.

Conceptually, simplifying the tax system can affect firm activity through two distinct channels. First, tax systems may be so complex that they are difficult to understand and interpret. Simplification may then help firms avoid costs related to misinterpreting the rules (Abeler and Jäger 2015; Zwick 2021). Second, tax complexity can give rise to administrative burdens that generate compliance costs. Complying with tax obligations requires firms to fulfill many reporting requirements, often in different languages (Amberger et al. 2025). Simplification can reduce such compliance costs and free up resources for core business activities. In this paper, we focus on the latter channel.

To study this channel, we leverage the EU's 2021 VAT reform, which simplified tax reporting for cross-border business-to-consumer (B2C) transactions while leaving other characteristics, such as tax rates or bases, unchanged. The cross-border VAT setting is well suited to studying firm responses to simplification for three reasons. Complexity-induced compliance costs are especially pronounced for VAT, and even more so for cross-border transactions (International Monetary Fund 2001; Harju et al. 2019; Mascagni et al. 2023). At the same time, VAT is a major revenue source for EU member states, where it accounts

¹For example, in 2025, the OECD convened a high-level symposium on "Simplifying for Success," bringing together government and business leaders to address regulatory complexity (OECD 2025b). Similarly, the 2024 Draghi report on EU competitiveness identified regulatory simplification as a key lever for restoring Europe's economic dynamism (European Union 2024).

for 15.7% of total revenues (European Commission 2022b).² In addition, the continued expansion of globalized online shopping through platforms such as Amazon, Temu, and Shein has led to a rise in cross-border B2C transactions (Mínguez and Minondo 2025).

In the EU, cross-border B2C transactions are generally taxed on a destination basis. This means that VAT is owed in the consumer's country, and the exporting firm has to report and remit it abroad.³ Consider a German firm selling goods online to households in France and Bulgaria. Before the reform, it had to register for VAT separately in France and in Bulgaria, filing returns under each country's rules, completing each set of forms, and often filing in the local language. The reform we study introduced two major simplification schemes, the One Stop Shop (OSS) and the Import One Stop Shop (IOSS) portals, that let exporting firms report and remit VAT for all destination countries through a single portal. The OSS is available to EU firms exporting to other member states, and the IOSS to non-EU firms exporting to EU consumers. Importantly, the change did not affect business-to-business (B2B) transactions. For the firm in our example, this means it no longer registers in France and Bulgaria at all. It now registers only in Germany and reports its French and Bulgarian sales through a single OSS return filed in its own language.⁴

The stated objective of the EU reform was to reduce compliance costs and, ultimately, stimulate business activity. We expect the reform to affect firms' cross-border activity along three margins. First, firms may pass some of the savings from lower ongoing compliance costs on to their customers (*price pass-through*), resulting in lower prices and greater demand. Second, simplifying VAT registration and reporting potentially frees up firm capacities and allows firms to allocate more resources to their existing activities (*resource freeing*), for example by directing human or financial resources toward international marketing or expanding the range of products sold. Third, a single portal for all cross-border sales reduces the fixed costs of expanding to an additional destination country (*geographic expansion*), as expansion no longer requires acquiring new country-specific knowledge, related for instance to language or documentation requirements, or engaging additional tax advisors. This could make exporting to additional countries more attractive and enable firms to reach new customers.

²Generally, consumption taxes are among the largest sources of tax revenue for governments worldwide. In the OECD, they account for roughly 30% of total tax revenue. Corporate income taxes represent much less, at 12% of total tax revenue in OECD countries (OECD 2024; Ghosh et al. 2025).

³Although consumers pay VAT as part of the purchase price, the exporting firm is responsible for registering, reporting, and remitting the tax to the authorities in the destination country.

⁴Figure 1 Panel (a) illustrates this change contrasting the pre- and post-reform reporting environment for intra-community B2C sales.

However, these positive effects on trading activity might not materialize if firms incur adjustment costs when adopting the new portal or have already optimized their existing systems under the prior regime, leaving little scope for further increases in sales activity. The reform’s potential to stimulate activity also depends on whether firms perceive it as genuinely reducing complexity. To obtain a first impression of how firms view the reform, we collect survey data through the German Business Panel (Bischof et al. 2025; Gassen and Muhn 2025; Winter et al. 2025). The survey evidence indicates that firms perceive the new portal as reducing VAT filing complexity, citing standardized procedures, efficiency gains, and the ability to file in their native language as key advantages.

To empirically study the effects of the simplified EU VAT system on firms, we rely on multiple high-frequency, firm- and transaction-level administrative data sets from Germany. Given its large trade volumes, Germany is one of the EU’s most active participants in cross-border commerce, so evidence from this setting is likely to carry external validity for other major economies. The data sets include tens of millions of observations and have not previously been available to researchers. Specifically, we combine VAT returns, foreign trade statistics, and customs data from before and after the reform.

In our first set of analyses, we assess the real effects of the OSS on within-EU sales activity of German firms. We employ a difference-in-differences (DiD) research design that compares the evolution of treated B2C EU-exports of German firms before and after the reform (first difference) with the corresponding evolution of unaffected German B2B EU-exports over the same period (second difference). Combining monthly foreign trade statistics at the transaction level with firm-level VAT-return data for the years 2018 to 2023, we estimate firm-level regressions and define a firm’s treatment status based on 5-digit industry codes.

We find that the number of cross-border transactions among treated firms increases significantly relative to control firms following the reform. The treatment effect is economically sizable, with treated firms exporting roughly 34% more transactions than control firms after the OSS. This equals approximately 4 additional transactions per firm-month relative to the pre-reform average. The increase in transaction counts is accompanied by no significant rise in sales revenues. We return to this gap when testing for price effects. We further find greater diversification in traded products, consistent with the resource-freeing channel, and an expansion in the number of destination countries served, consistent with geographic expansion and firms no longer needing to register separately in every destination. To confirm that these results are not driven by pre-existing trend differences between

treatment and control firms, we plot coefficients from a dynamic DiD specification. Pre-reform coefficients are close to zero and statistically insignificant, supporting the parallel trends assumption, while the treatment effect builds gradually following the reform.

We test the robustness of these results using different treatment and control groups. The new portal is not available for B2C transactions by EU firms to non-EU countries, allowing us to compare German B2C firms that primarily export to EU member states with those that primarily export to non-EU countries. Using this design, we again find approximately 3 additional transactions per firm-month, similar to the estimate from the B2B control group. The effects on sales, product scope, and destination countries are likewise (qualitatively and quantitatively) similar.

To shed light on the mechanisms underlying the positive real effects on business activity, we examine the three channels outlined above. First, *price pass-through* could explain why transaction counts increase more than sales revenues: firms may pass part of their compliance cost savings on to customers through lower prices. Consistent with this mechanism, mean firm-product prices decline by 2.6 % after the reform, while mean product weights do not change significantly, ruling out a shift toward cheaper and lighter goods. Second, the results support the *resource-freeing channel*. Product scope increases following the reform, particularly among firms that already exported to multiple countries. These firms had geographically diversified operations and may therefore have redirected administrative capacity released by simplified reporting toward product diversification rather than further geographic expansion. This pattern is consistent with the portal reducing the ongoing compliance burden of cross-border sales. Third, our findings are consistent with the *geographic expansion channel* firms increase the number of destination countries they serve, with similar percentage increases among firms with one or multiple pre-reform destinations. The effect is concentrated among small and medium-sized firms, consistent with lower destination-specific compliance costs being particularly valuable for firms with limited administrative resources. Our survey further supports this mechanism: firms identify the ability to file in their native language as a key advantage of the new system, highlighting the importance of country-specific knowledge and language for market entry.

In a second set of analyses, we turn to the second part of the reform, the IOSS. It extends the same single-portal simplification to non-EU firms exporting to consumers in the EU, but unlike the OSS it applies only to transactions valued below EUR 150. Recalling our example from earlier, we now apply it to a Chinese firm selling to consumers in Germany and France. Under the reform, it can use the IOSS to report all transactions below

EUR 150 through a single portal, whereas previously it had to register and report in each country separately.⁵ Such eligibility thresholds are a common feature of regulatory and tax policy. Prominent examples include simplified accounting (e.g., Bernard et al. 2018) and tax regimes (Aghion et al. 2025), both of which rely on size-based (e.g., revenue) cutoffs.⁶

In the IOSS setting, the eligibility threshold gives firms a second way to respond, beyond the real activity changes we documented for the OSS. Firms can engage in strategic reporting to access the regime’s benefits without changing their underlying business activity, for instance by misdeclaring transaction values, or by splitting a shipment worth more than EUR 150 into several smaller parcels that each fall below the cutoff. The two responses differ in what they imply for the policy. A real effect reflects the policy working as intended, whereas strategic reporting is an unintended consequence that can erode the tax base. Distinguishing the two is therefore important, yet in most settings and data they are difficult to separate. The IOSS setting lets us do so. We first study the reform’s real effects and replicate our earlier findings, and we then provide evidence that distinguishes real responses from reporting responses around the threshold.

For the IOSS analysis, we use the transaction-level customs data introduced above, comprising roughly 162 million observations for the years 2018 to 2023 across transactions valued between zero and EUR 1,500. The real effects we find in this setting are considerably larger than under the OSS, which suggests that firm responses around the eligibility threshold contribute to the overall pattern beyond the baseline increase in activity. We therefore zoom into firm responses around this threshold.

Following the logic of a bunching analysis, we plot the number of transactions in narrow bins around the EUR 150 threshold, separately for pre- and post-reform periods (Burgstahler and Dichev 1997). In pre-reform years, we detect modest bunching below the EUR 150 threshold, which is expected because transactions under this value are tariff-free both prior to and after the simplification reform. Otherwise, the distribution appears smooth, with no indication of a missing mass above the threshold. After the new portal became accessible, we observe a pronounced hole just above the EUR 150 threshold and a corresponding increase in transactions below EUR 150, indicating that firms shift their transactions below the threshold. Quantifying this shift following Bernard et al. (2018),

⁵Figure 1 Panel (b) illustrates this change, contrasting the pre- and post-reform reporting environment for extra-community B2C sales.

⁶Further examples include the U.S. Qualified Small Business Stock exclusion under Section 1202, which grants tax benefits only to firms whose gross assets fall below a specified threshold, and international regulations such as the Global Minimum Tax and country-by-country reporting which apply only to large multinational firms above certain revenue thresholds, exempting smaller firms from these requirements.

we estimate that after the reform, B2C transactions became 34% to 49% more likely than B2B transactions to be declared below EUR 150.

In additional tests, we confirm that this response is driven by the simplification reform. In placebo tests using B2B transactions, for which the portal cannot be used, we document no such pattern. Furthermore, the tariff exemption cutoff cannot explain the pattern, as the same exemption was in place before the reform yet produced no comparable missing mass above the threshold.⁷ Splitting the sample by Germany’s two most important source countries, the U.S. and China, we find the same distributional pattern for both, though it is somewhat more pronounced for China.

We next assess whether real or reporting responses lead to this pattern. The real effects explanation cannot fully rationalize our findings. While expanded supply of lower-priced goods or marginal price reductions could generate the excess mass below the threshold, neither can explain the pronounced decline in transactions above it, particularly the missing mass that extends well beyond the threshold. This pattern instead points to misreporting or parcel splitting.

To test specifically for parcel splitting, we examine transaction weights. If firms divide a transaction into several lighter parcels to bring each below the threshold, the average parcel should become lighter at a given declared value, a change that a real price reduction would not produce. We therefore plot the average transaction weight (i.e., parcel weight) in narrow price bins around the EUR 150 threshold. We find that transaction weight decreases in every price bin after the portal becomes available, whereas we observe no comparable decline for B2B transactions. Concretely, a transaction with a true value of EUR 170 split into two lighter parcels now appears as two sub-threshold transactions, changing how the transaction is reported without changing the underlying business activity.⁸ Such parcel splitting constitutes an unintended reporting consequence of the reform, as firms misreport values and report parcels separately that should be declared as a single transaction in order to qualify for the simplified portal.

Our paper contributes to three streams of literature. First, we add to the literature on tax and reporting complexity. Prior research shows that complex tax and reporting systems impose substantial economic costs, reducing refund uptake (Zwick 2021), foreign direct

⁷As a further check, we examine goods subject to tariffs separately from those that are not, and find consistent results for both.

⁸This parcel splitting is more plausibly driven by firm behavior than by consumers. Since the threshold for using the portal applies to the net price, while consumers pay the gross price under the IOSS regime, the cutoff is not salient to consumers, so it is unlikely that they strategically split their orders to fall below the threshold.

investment (Esteller-More et al. 2021; Euler et al. 2024), employment growth (Brown et al. 2025), and the effectiveness of tax policy (Amberger et al. 2025), while also deteriorating firms' information environments and disclosure quality (Brown et al. 2023). Related work examines how firms cope with complexity within a given regulatory environment, for instance by using tax preparers (Zwick 2021), developing in-house accounting expertise (Chychyla et al. 2019), or relying on taxpayer assistance centers (Armstrong and Glaeser 2026). We extend this literature by studying firm responses to a direct regulatory intervention that reduces complexity. Because the reform lowers compliance costs directly, we can show that simplification delivers its intended effects.

Second, we contribute to the literature on the real effects of reporting regulation and firms' strategic responses to reporting incentives. Prior research primarily examines the consequences of additional reporting requirements, such as country-by-country reporting mandates (De Simone and Olbert 2022; Nessa et al. 2025) and other measures aimed at curbing tax avoidance (Bilicka et al. 2025). In contrast, we study a reform that reduces rather than expands reporting obligations and document that simplified reporting translates into real economic responses, providing evidence that lower compliance costs can stimulate business activity. Beyond documenting these direct real effects, we also shed light on how reporting incentives shape firm behavior when simplified regimes are combined with eligibility thresholds, a common design feature of tax policies. In our second setting, we find that when simplification is paired with such a threshold, firms respond along the real margin but also engage in strategic reporting behavior to access the regime's benefits (Aghion et al. 2025). We provide evidence that distinguishes these two responses, using transaction weight to isolate parcel splitting from real price adjustment. Documenting real and reporting responses together in a single setting speaks directly to the interaction that Lester and Olbert (2025) call on accounting researchers to examine. Our findings show that reporting incentives can undermine the objectives of simplification, connecting to a broader literature on how reporting design induces strategic firm behavior (De Simone et al. 2020; Williams and Williams 2021; Kelley et al. 2024).

Third, we contribute to the literature on how VAT design shapes firm behavior. Prior work examines how VAT rates and adoption decisions affect investment (Jacob et al. 2019; Chen et al. 2023), payout behavior (Jacob et al. 2024), cross-border revenue allocation (De Simone and Olbert 2025), and product mix (Ghosh et al. 2025), while a separate compliance literature shows that VAT design shapes the extent to which firms report transactions (Pomeranz 2015; Naritomi 2019). We bridge the gap between both streams by

studying how the design of VAT reporting regulations, specifically their complexity, affects real cross-border activity. Using granular, high-frequency transaction-level data, we provide novel evidence that simplifying reporting obligations expands real economic activity and that eligibility thresholds in the simplified regime induce strategic reporting alongside real responses. This evidence also responds to a call by Dyreng et al. (2025) to study non-income taxes, an area identified as underexplored yet increasingly important given the global shift toward alternative revenue sources.

2 Institutional Details and Conceptual Framework

2.1 Simplifying VAT Reporting

VAT is a broad-based consumption tax levied on most goods and services in over 170 countries worldwide.⁹ Unlike the U.S. retail sales tax, which is collected only at the final point of sale, VAT is collected at each stage of the supply chain. Firms charge VAT on their sales (output VAT), reclaim VAT paid on their purchases (input VAT), and remit the difference to the tax authority.¹⁰ For cross-border B2C transactions, VAT is generally assessed on a destination basis, meaning that the tax is owed in the country where the consumer resides and at that country’s applicable rate.¹¹ Critically, the obligation to report, collect, and remit the VAT falls on the selling firm rather than the consumer.

We study firms exporting goods to consumers in EU member states. Our setting distinguishes between EU firms selling to consumers in other EU member states (intra-community sales) and non-EU firms selling to consumers in the EU (extra-community imports). In both cases, the selling firm is responsible for registering for VAT, reporting transactions, and remitting the collected tax to the destination country. Prior to the reform, this created a highly fragmented compliance environment. A firm selling to consumers across multiple EU member states had to register for VAT separately in each destination country, navigate the distinct reporting systems of each national tax authority, and file returns according to each country’s specific requirements. Given that the EU comprises 27 member states, each with its own registration procedures, filing deadlines, return formats,

⁹For a comprehensive overview of VAT systems, see Keen (2013) and OECD (2024).

¹⁰An exemption applies to small businesses. For example, in Germany, small businesses whose sales in the previous year did not exceed EUR 17,500 and whose current sales do not exceed EUR 50,000 are exempt from VAT.

¹¹The destination principle applies to firms exceeding an annual threshold of EUR 10,000 in total cross-border sales and thus to the majority of firms. Firms below this threshold follow the origin principle.

language requirements, and VAT rates ranging from 17% to 27%, these firms faced considerable complexity. Even modest expansion into additional markets required firms to manage an entirely new set of administrative requirements for each country entered.

To reduce this complexity, the EU introduced the VAT e-commerce package in July 2021. The reform established two simplification mechanisms that allow firms to report and remit VAT for multiple jurisdictions through a single online portal in one EU member state: the *One Stop Shop* for intra-community sales and the *Import One Stop Shop* for extra-community imports. Figure 1 and Table 1 illustrate the regulatory changes introduced by these portals.

The OSS is available to EU firms selling to consumers in other EU member states and can be used for B2C transactions of any value. Rather than registering separately in each destination country, firms register once in their home member state and submit monthly returns covering all their intra-EU B2C sales. The portal calculates the VAT owed to each destination country based on the applicable local rate, and the firm remits the total amount to its home country's tax authority, which then distributes the revenues to the respective member states. Firms typically select their home country as the member state of registration, allowing them to report in their own language. Registration for the OSS is voluntary. Firms that do not use the OSS must continue to register separately in each country where they sell to consumers. The reform thus creates a strong incentive to adopt the OSS, particularly for firms operating across multiple countries, as it eliminates the need to manage different national VAT systems.

To illustrate the scope of the change, consider a German firm that sells products to final customers in France and Bulgaria. Prior to the reform, this typically required either engaging an external tax advisor in each destination country or building substantial in-house capacity to manage two separate VAT registrations, each with its own deadlines, forms, and language requirements. Under the reformed system, the firm registers once and reports all cross-border B2C transactions through a single standardized portal in German. If it subsequently decides to enter a new market, e.g., the Netherlands, no additional VAT registration or country-specific setup is required.

Our second setting, the IOSS, provides an analogous simplification for firms located outside the EU that sell goods to EU consumers. A critical feature of the IOSS is that it applies only to transactions valued at EUR 150 or less.¹² Under the IOSS, VAT is charged

¹²We use the term *transaction* interchangeably with the legal term *consignment*. Following the definition in the legal text of the VAT e-commerce package, a consignment (transaction) refers to goods packed together and dispatched simultaneously by the same supplier to the same consignee under a single transport con-

to consumers at the point of sale, and firms report and remit the collected VAT monthly to their chosen EU member state. The EUR 150 threshold creates a discrete change in reporting requirements. Transactions for which firms can use the IOSS benefit from streamlined reporting, faster customs clearance, and a reduced administrative burden. Transactions above the threshold, by contrast, require full import VAT and customs declarations and the associated procedural complexities.¹³ The EUR 150 threshold coincides with the tariff exemption threshold. Because the exemption predates the reform and remained unchanged, any change in behavior around the threshold after the reform is attributable to the IOSS rather than to the tariff exemption.

Alongside the IOSS, the deemed supplier provision was introduced, under which online marketplaces facilitating sales by non-EU sellers are treated as the supplier for VAT purposes and bear the responsibility for collecting and remitting the tax. Crucially, this provision applies to transactions both above and below EUR 150, so it does not generate differential incentives around the threshold and is therefore unlikely to confound the behavioral responses we document. The reform also abolished the former EUR 22 VAT exemption for low-value imports, meaning that all goods imported into the EU became subject to VAT regardless of value. To avoid confounding effects from this change, we exclude transactions valued below EUR 30 from our analyses.

Importantly, neither the OSS nor the IOSS changes the VAT burden itself, as the applicable rate remains that of the destination country. The application and amount of tariffs are also unaffected. Both portals apply exclusively to B2C transactions and do not change the reporting obligations for B2B sales, which we leverage as a control group in our empirical analysis.

2.2 Conceptual Framework

Cross-border VAT compliance imposes substantial costs on firms. Prior to the reform, firms selling to consumers across multiple EU member states had to maintain separate VAT registrations in each destination country, comply with jurisdiction-specific documentation and filing requirements, and often engage external tax advisors to navigate foreign-language procedures, generating significant compliance costs beyond the tax liability itself

tract. The EUR 150 threshold applies to the total transaction value rather than to individual parcels within a transaction, excluding transport and insurance costs, and VAT.

¹³Tariffs apply to some extra-community imports. In general, they apply only to transactions above EUR 150. However, whether a specific product is subject to customs duties and the applicable rate depends on the type of product.

(James 1990). To put this in perspective, EU businesses are estimated to spend a combined EUR 204 billion annually on tax compliance. Among the various taxes, businesses regard VAT and corporate tax as imposing the highest compliance burden, with the majority of smaller enterprises required to outsource these tasks to specialized firms (European Commission 2022a).

Compliance costs in this cross-border setting take two distinct forms. First, there are fixed costs associated with entering each new destination market, including VAT registration, acquiring country-specific knowledge, and setting up the administrative infrastructure to meet local reporting requirements. These costs must be incurred upfront and are largely independent of transaction volume, functioning analogously to the fixed costs of market entry in the trade literature (Melitz 2003). Second, there are variable costs arising from the ongoing obligation to report and document individual transactions separately for each jurisdiction. Both types of costs fall disproportionately on smaller firms and on firms with limited prior cross-border experience, for whom the per-unit compliance burden is highest. Prior literature has found that such reporting and compliance obligations reduce firm activity across distinct settings (Zwick 2021; Amberger et al. 2025; Brown et al. 2025).

By enabling firms to report VAT for all destination countries through a single online portal and thus simplifying the tax reporting and remitting process, the reform directly reduces both the fixed and variable components of cross-border compliance costs. Consistent with prior empirical work showing that reduced tax and regulatory costs stimulate entrepreneurship and business activity (Klapper et al. 2006; Djankov et al. 2010; Armstrong and Glaeser 2026), we expect the simplified reporting regime to increase overall business activity in the form of transaction frequency. Reduced compliance costs lower the effective cost of serving foreign markets, and firms may pass at least part of these savings on to consumers in the form of lower prices (*price pass-through*), stimulating demand.

Beyond the price pass-through channel, reductions in ongoing compliance costs can free up human and financial resources that firms can use for revenue-generating activities (*resource freeing*). Firms adopting the simplified regime no longer need to devote the same administrative capacity to managing jurisdiction-specific reporting obligations and may instead use these resources for international marketing, product development, or expanding the range of goods offered in existing markets. We therefore expect firms to respond along the product scope margin, especially firms that already operate across several destination countries and can use the simplified reporting system to deepen existing trading activities.

We further expect the reform to stimulate *geographic expansion*. By eliminating separate country-specific VAT registration and replacing foreign-language filing requirements with a standardized procedure in the firm’s home language, the reform reduces the fixed costs of entering additional destination markets. This effect should be particularly relevant for firms with limited administrative resources, such as small and medium-sized firms, for whom destination-specific compliance requirements are likely to be most binding.

Several frictions could dampen or delay these effects. Firms face adjustment costs when adopting a new compliance system, and those that had previously invested in optimizing their compliance structures under the prior regime may have limited scope for further efficiency gains. If the reform lacked sufficient salience or was not perceived as meaningfully simplifying the system, take-up may have been limited.

In contrast to the OSS, the IOSS restricts access to the simplified regime to transactions valued at EUR 150 or less, a design feature shared by many other regulatory simplification measures (see Section 1). This threshold introduces a qualitatively different set of incentives. Firms seeking to benefit from the portal’s compliance cost savings may respond not only along real margins, for instance by expanding the supply of lower-priced goods or making marginal price adjustments to bring transactions below the cutoff, but also through strategic reporting behavior such as understating declared transaction values or splitting transactions into multiple parcels to qualify for the portal (Aghion et al. 2025).

2.3 Firm Survey

To better understand how firms perceive the reform, we conduct a survey among German firms. The survey was administered by the German Business Panel (Bischof et al. 2025; Gassen and Muhn 2025; Winter et al. 2025) during spring and summer 2025, roughly four years after the implementation of the EU VAT e-commerce reform. This timing ensures that firms had sufficient experience to form well-grounded views. Responses were collected anonymously through an online questionnaire administered via Qualtrics.

The survey evidence speaks directly to the frictions discussed above, namely whether the reform was sufficiently salient and whether firms perceived it as genuinely simplifying the system. We ask a total of 3,276 firms whether they engage in cross-border B2C sales within the EU, of which about 21% report having done so in recent years. These firms receive additional questions about their VAT declaration practices and OSS adoption. Among OSS users, 63% indicate that VAT filing has become simpler under the OSS, with standardized reporting procedures, efficiency gains, and the ability to file in their native language

cited as the main advantages. These responses suggest that, among adopting firms, the reform was both salient and perceived as delivering meaningful simplification, mitigating concerns that limited awareness or skepticism about the reform’s benefits would suppress a real response. Finally, we asked whether the OSS prompted firms to expand their sales to additional EU member states. Notably, 11% reported doing so as a direct result of the reform, providing preliminary survey-based evidence consistent with the geographic expansion mechanism discussed above. Key survey results are summarized in Figure 2.

3 Effect of Simplified Reporting on Business Activity

3.1 Empirical Strategy

To estimate the real effects of simplified VAT reporting on firm-level business activity, we exploit the EU-wide implementation of the OSS in July 2021. Because the OSS applies to transactions of any value and imposes no eligibility threshold, it creates no incentive for strategic reporting responses. We therefore interpret any response we detect in this setting as a real change in business activity. We employ a difference-in-differences research design. Our baseline specification compares treated B2C firms to control B2B firms selling to EU member states before and after the reform. Since B2B transactions are not eligible for the OSS, the evolution of B2B exports provides a benchmark against which to assess changes in intra-EU B2C sales attributable to the reform. We note that not all firms in the treatment group adopt the OSS despite its simplification benefits, and our estimates should therefore be interpreted as intent-to-treat effects. Official sources suggest that adoption among German firms was substantial.¹⁴

We employ a Poisson pseudo-maximum likelihood (PPML) estimator, which is well suited for count-based outcome variables and handles the prevalence of zeros in trade data more effectively than log-linear alternatives (Santos Silva and Tenreyro 2006; Cohn et al. 2022; Chen and Roth 2023; Wooldridge 2023). Specifically, for firm i and month t we estimate the following specification:

$$SALESVariable_{it} = \beta_1(TREAT_i \times POST_t) + \mathbf{W}_{it} + \alpha_i + \alpha_t + \varepsilon_{it} \quad (1)$$

¹⁴According to the German Federal Government, 23,638 businesses had already registered for the OSS by 6 August 2021, only weeks after the reform’s introduction (German Federal Government 2021), and the European Commission documents a continued increase in OSS registrations across the EU, describing the VAT e-commerce package as a successful reform (European Commission 2025).

where $SALESVariable_{it}$ captures a firm’s monthly cross-border business activity, measured alternatively as the number of transactions (*Count*), total sales volume (*Sales*), the number of distinct goods traded cross-border (*Goods*), and the number of distinct destination countries (*Countries*). $TREAT_i$ is an indicator variable equal to one for firms exposed to the OSS reform and zero otherwise. In the baseline specification, $TREAT_i$ equals one for B2C firms selling within the EU and zero for B2B firms trading within the EU. Following prior literature, we classify retailers and wholesalers primarily engaged in B2C sales as the treatment group (De Simone and Olbert 2025).¹⁵ $POST_t$ equals one for the post reform period from July 2021 to December 2023 and zero otherwise. Firm fixed effects (α_i) absorb time-invariant heterogeneity across firms, while year-month fixed effects (α_t) capture aggregate shocks common to all firms in a given period. Standard errors are clustered at the firm level. In some specifications, we additionally include a vector of control variables (W_{it}). These include pre-treatment average cross-border transaction counts and/or sales volume that we interact with a linear time trend (specification labeled as *cross-border trade controls*), or pre-treatment values of the number of employees and total firm sales interacted with a linear time trend (specification labeled as *firm-level controls*). We use pre-treatment control variables to avoid bad controls. Our results are robust across specifications with different structures of control variables. The coefficient of interest is β_1 , which captures the average change in business activity for treated firms relative to control firms following the reform. A positive and significant β_1 is consistent with the reform stimulating cross-border business activity.

We choose the B2B control group for two reasons. First, both groups operate within the same intra-European market, holding the broader demand environment constant. Second, B2B and B2C transactions are administratively distinct, so the treatment and control groups are clearly delineated with no risk of overlap. Consistent with a valid control group, pre-reform trends in business activity are similar across the two groups as we show in Section 3.3. A potential concern is that the reform indirectly affects B2B firms if expanded B2C activity substitutes for distribution through wholesalers and retailers, thereby reducing demand for B2B transactions. We do not view this as a material threat to identification. The substitution channel would require foreign retail consumers to shift toward direct cross-border online purchases from German firms rather than purchasing through domestic intermediaries that import their products from German firms, a margin that is less immediate and less likely to be driven by the OSS reform. Moreover, if firms were

¹⁵See Appendix Table A.3 for the industry descriptions and codes included in the treatment group.

to shift their business model from B2B toward direct-to-consumer sales following the reform, this would attenuate our estimates rather than inflate them, since treatment status is assigned based on pre-reform industry classification and a firm switching toward B2C post-reform would be counted as a control firm expanding its activity, pushing the estimated treatment effect downward.

That said, we acknowledge that B2C and B2B firms may differ systematically in their business models, customer structures, and trading behavior in ways not fully captured by firm fixed effects and pre-treatment controls. We therefore additionally estimate an alternative specification that restricts the comparison to B2C firms only. Specifically, we compare German B2C firms primarily exporting to EU member states with those primarily exporting to non-EU countries. Since extra-EU transactions are not eligible for the OSS, this setup exploits variation in reform exposure within the same customer segment and industry classification, thereby addressing concerns about systematic differences between B2C and B2B firms. Firms with more than 60% of their exports directed to EU destinations are classified as treated, while firms with less than 40% of exports to the EU are classified as the control group.¹⁶ The results remain qualitatively and quantitatively consistent across both specifications, supporting the robustness of our main findings.

3.2 Data, Sample and Descriptives

We combine two German administrative data sources to test whether simplified tax reporting affects firm-level business activity. The foreign trade statistics record all cross-border transactions by German firms at the transaction level, providing detailed information on transaction values, product codes, and destination countries.¹⁷ The VAT return data contain yearly firm-level information on total taxable turnover and firm characteristics including industry sector and number of employees.¹⁸

Appendix Table A.2 reports our sample selection procedure. We begin by merging the foreign trade statistics with the VAT return data using a firm identifier, yielding a firm-transaction level dataset covering the years 2018 to 2023.¹⁹ We exclude observations with missing industry codes, as we identify treatment and control groups based on the five-digit

¹⁶Appendix Table A.7 reports summary statistics for this alternative sample. The gap between the two thresholds is intended to avoid misclassifying firms with mixed export destinations.

¹⁷This dataset is provided by the Research Data Center of the Federal Statistical Office and Statistical Offices of the Federal States, DOI: <https://doi.org/10.21242/51911.2024.00.05.1.1.0>.

¹⁸This dataset is provided by the Research Data Center of the Federal Statistical Office and Statistical Offices of the Federal States, DOI: <https://doi.org/10.21242/73311.2023.00.01.1.1.0>.

¹⁹The VAT return data are available only for the pre-treatment period from 2018 to 2021.

industry classification. We further exclude import transactions, extra-EU transactions, and transactions involving Great Britain, as the Brexit referendum coincides with our sample period. We additionally exclude transactions subject to excise duties as these are governed by a separate tax regime and transactions already covered by the predecessor Mini-One-Stop-Shop (MOSS) regime.²⁰

Since we are interested in the effects of simplified reporting on firm-level business activity, we aggregate the transaction-level data to a firm-by-year-month panel. Our variables of interest are the number of transactions (*Count*), total sales volume (*Sales*), the number of distinct goods sold (*Goods*), and the number of distinct destination countries (*Countries*), each by firm and year-month. This sample selection procedure results in 8,6 million observations for 348,004 firms.²¹

Table 2 reports summary statistics for the variables used in our main analyses. Panel A reports descriptive statistics for treated B2C firms and Panel B for untreated B2B control firms. On average, a German B2C firm conducted 12 cross-border transactions per month, compared to 21 for B2B firms. B2C firms typically trade with two partner countries and often sell only a single type of good, while B2B firms generally operate across more markets and offer a wider product range. These differences in levels are expected given the distinct nature of B2C and B2B trade. They do not affect the validity of our DiD design, which relies on parallel pre-reform trends rather than comparable levels.²²

Appendix Table A.5 provides an overview of the main destination countries and most commonly traded goods for cross-border B2C sales by German firms in the pre-reform period. Panel A shows that the most frequent destination countries are geographically close to Germany, with Austria appearing most often, followed by the Netherlands, France, and Poland. Austria's prominence likely reflects the shared language and cultural proximity between the two countries, which can lower the barriers to cross-border trade. Panel B reports the most frequently exported product categories.²³

²⁰The Mini-One-Stop-Shop was the predecessor to the OSS, introduced in 2015 for cross-border B2C sales of digital services. We exclude firms that had already registered for MOSS prior to the reform to ensure a clean pre-reform baseline. See Appendix Table A.4 for details.

²¹Note that not all firms have cross-border transactions in each month of our sample period.

²²In untabulated tests, we confirm that treated and control firms exhibit similar pre-treatment trends across all dependent variables.

²³Full product group descriptions are provided in Appendix Table A.6.

3.3 Results

Table 3 presents the results of estimating Equation 1 to test whether simplified VAT reporting increases cross-border business activity. Panel A reports estimates including only fixed effects, Panel B adds cross-border trade controls, and Panel C adds firm-level controls.

Across all three specifications, we find similar positive and statistically significant effects on the overall number of cross-border transactions, consistent with the reform increasing cross-border business activity. Our preferred estimate in Panel B implies that B2C firms increase their number of cross-border transactions by approximately 34% relative to control firms following the reform.²⁴ Relative to the pre-reform average of 12 transactions per month, this corresponds to an increase of approximately 4 transactions per month.

For sales volume, we find no statistically significant increase. However, the coefficients in Panels A to C are positive and similar in magnitude, yet considerably smaller than the estimated effects on transaction counts. The weaker response of sales volume relative to transaction frequency suggests that average transaction values may have declined following the reform, consistent with firms passing on compliance cost savings to consumers. We test this interpretation in Section 3.4.

We next examine whether simplified reporting increases the number of distinct goods traded cross-border. We find a positive and statistically significant effect of approximately 5.4%, suggesting that firms expand their product scope following the reform. This is consistent with the resource-freeing mechanism, whereby reductions in variable compliance costs free up firm resources that can be redirected toward marketing additional products.

Finally, we examine whether the reform reduced the fixed costs of entering new destination markets and thereby increased the number of countries served. We find a positive and significant effect on the number of destination countries across all specifications. Our preferred estimate implies an increase of approximately 2.0%, suggesting that firms broaden their geographic reach following the introduction of the OSS.

To shed light on the dynamics of the treatment effect and to confirm that the results are not driven by pre-existing trend differences between treatment and control firms, Figure 3 presents event-study plots based on our preferred specification, replacing the *Post* indicator with separate year-quarter indicators. Aggregating the monthly firm-level panel to the firm-quarter level allows us to display the full post-reform period and trace how firm responses evolve over time following the introduction of the OSS. Across all panels, the coefficients for the pre-reform quarters are close to zero, supporting the parallel trends

²⁴The underlying coefficient estimate is calculated as $(\exp(0.295) - 1) \times 100$.

assumption underlying our DiD design. We observe a temporary spike around the fourth quarter of 2020 (-2), which we attribute to the COVID-19 pandemic, as this was the first Christmas during which EU citizens faced lockdown restrictions, generating a sudden but short-lived increase in cross-border transactions. The spike is absent in the product scope and destination countries panels, consistent with it reflecting a transitory demand shock rather than a structural change in firm behavior.²⁵

Following the introduction of the OSS, cross-border business activity increases gradually across all four outcomes. The effect on transaction counts becomes statistically significant two quarters after the July 2021 implementation and continues to grow through the end of the sample period. The corresponding effect on sales volume follows a similar trajectory but remains more modest in magnitude and noisier, suggesting that price effects partially offset the volume gains. The effects on destination countries and product scope are smaller in magnitude but similarly gradual, becoming statistically significant within the first year following the reform and continuing to grow thereafter. The delayed onset across all outcomes is consistent with firms implementing major operational changes at the start of a new fiscal year rather than mid-year, including both the adoption of the new reporting regime and subsequent business expansions into new markets or product ranges.

Although the observed patterns are consistent with firms expanding their cross-border activity in response to the reform, we address the previously discussed concern that the results may reflect systematic differences in the business models, customer structures, or trading behavior of B2C and B2B firms rather than the OSS itself. To assess this, we re-estimate our main specification restricting the comparison to B2C firms only. Specifically, we compare B2C firms primarily exporting to EU member states with B2C firms primarily exporting to non-EU consumers, for whom extra-EU transactions are not eligible for the OSS and reporting obligations were therefore unchanged by the reform. Under this design, a general divergence between B2C and B2B firms would not generate a positive treatment effect, since both the treatment and control groups consist of B2C firms.

Table 4 reports the results and Appendix Figure A.1 reports the corresponding event study graphs. Across all specifications, the coefficient on the interaction term remains positive and statistically significant for transaction counts, product scope, and destination countries. The sales volume coefficient, while positive, is not statistically significant in this specification, consistent with the more modest and noisier sales volume effects already

²⁵We are not concerned that the COVID-19 pandemic confounds our main results. No major border closures for goods were imposed between the countries in our sample around the time of treatment. In addition, results are unchanged when we include an indicator variable for lockdown months (unreported).

observed in the baseline results. The magnitudes are broadly comparable to the baseline, though smaller on average, consistent with the treatment intensity design in which firms are classified based on the share of exports directed to EU destinations. In summary, these results indicate that the increase in cross-border activity following the introduction of the OSS is unlikely to be driven by structural differences between B2C and B2B firms.

3.4 Mechanisms

We next examine heterogeneity in the business activity response across groups of firms to shed light on the mechanisms underlying the increase in cross-border activity. We follow our conceptual framework and test the following channels: price pass-through, resource freeing, and geographic expansion.

We first test whether the increase in business activity is partly driven by firms passing lower compliance costs through to consumers in the form of *reduced prices*, which in turn stimulates demand. A first indication of price adjustments emerges directly from the baseline estimates in Table 3. The increase in transaction counts is considerably larger than the increase in total sales volume, suggesting that the average transaction value, and thus the average price, may have declined following the reform. To assess this directly, we disaggregate the sample to the firm-product-by-year-month level, where each observation represents the average price of a given product charged by a given firm in year-month t . We then regress this average firm-product price on the interaction term, including firm-product and year-month fixed effects, to identify within-firm, within-product price changes attributable to the reform.

Panel A of Table 5 reports the results. In Column (1), we find a statistically significant decline in the average product price following the reform, consistent with firms passing at least part of their compliance cost savings on to consumers. However, this price decline could alternatively reflect a compositional shift in the set of products traded, for instance, if firms begin offering lower-value goods as cross-border trade becomes less costly. To rule out this explanation, Column (2) replicates the analysis using average transaction weight as the outcome variable. Since weight is correlated with product value, a significant change in average weight would suggest a shift toward lower-value goods rather than a real price reduction within existing products. We find no significant change in average transaction weights, indicating that the observed price decline reflects a within-product pricing adjustment rather than a compositional shift in traded goods.

We next test the *resource freeing* channel, which predicts that reductions in variable

compliance costs free up firm resources that can be reallocated toward expanding activity within existing markets. To test this, we split the sample based on the number of destination countries served prior to the reform, distinguishing firms already active across multiple markets from those exporting to only a single destination. For firms already operating across multiple markets, further geographic expansion may be less central margin of adjustment, because these firms had already established geographically diversified operations. Instead, they may use the released administrative capacity to broaden the range of products they sell. Panel B of Table 5 reports the results. While transaction counts increase significantly for both groups of firms, the increase in the number of distinct goods traded is substantially larger among firms that already exported to multiple countries before the reform. This pattern is consistent with the interpretation that reductions in ongoing compliance costs free up resources that are redeployed toward product scope expansion and deeper engagement within existing markets.

Finally, we test whether the increase in destination countries is particularly pronounced for firms that previously faced the highest fixed costs of *geographic expansion*. We return to the sample split based on the number of destination countries served prior to the reform. Firms exporting to a single destination country prior to the reform faced high fixed entry costs, as each additional market required separate VAT registration and navigation of foreign-language filing requirements. Panel B of Table 5 reports the results. We find that the percentage increase in destination countries is similar for firms that exported to a single destination and firms that already exported to multiple countries before the reform. This suggests that the geographic expansion effect does not strongly depend on firms' prior geographic scope. However, the response differs by firm size. Panel C shows that small and medium-sized firms significantly expand the number of destination countries served, whereas large firms show no significant geographic response. This pattern is consistent with compliance cost reductions being particularly valuable for firms with more limited administrative resources. Our survey evidence further corroborates the role of country-specific knowledge and language as key components of entry costs, with firms citing the ability to file in their native language as a key advantage of the new system (see Section 2.3).

4 Strategic Responses to Simplified Reporting

4.1 Empirical Strategy

The OSS analysis establishes that regulatory simplification increases real cross-border business activity. A common design feature of simplified regimes in tax and accounting regulations is the use of eligibility thresholds that restrict access to qualifying firms or transactions (Bernard et al. 2018; Aghion et al. 2025). The IOSS follows this design: it mirrors the OSS in its core functionality, enabling non-EU firms to report and remit VAT through a single portal, but restricts access to transactions valued at EUR 150 or less. This threshold allows us to examine whether the simplification effect documented under the OSS extends to a threshold-based regime and, more importantly, whether the threshold itself generates behavioral responses beyond those observed under the OSS. The transaction value threshold may create incentives for firms to respond not only along real margins, as in the OSS context, but also along the reporting margin, by misrepresenting declared transaction values or splitting parcels to qualify for the simplified portal. The IOSS therefore provides a setting in which real and reporting responses may occur simultaneously, and in which distinguishing between the two is important.

The empirical analysis draws on transaction-level customs data that we describe in Section 4.2. This additional administrative dataset differs from the datasets used in the OSS analysis in two important aspects. The dataset does not contain firm identifiers, preventing aggregation at the firm level, and records transactions at annual rather than monthly frequency.

We proceed in three steps, all of which draw on this dataset. First, we employ a DiD design that compares the evolution of eligible B2C imports with ineligible B2B imports before and after the introduction of the IOSS to confirm that simplification increases business activity for transactions below the threshold. Since the dataset does not contain firm identifiers, the transaction-level data lack the panel structure necessary to include unit-level fixed effects. We therefore aggregate to the product-country-year level, which allows us to include product-country fixed effects that absorb time-invariant differences in trade volumes across product-country pairs, such as differences in product demand or bilateral trade relationships, alongside year fixed effects that capture time-varying shocks. We restrict the sample to transactions valued between EUR 30 and EUR 150. Given that the reform took place in July 2021, we drop 2021 from the sample to avoid contaminating the pre- and post-reform periods with a partial treatment year. Finding a positive effect for

eligible B2C transactions below the threshold would confirm that the simplification effect documented in the OSS setting extends to the IOSS context.

Second, we examine whether the eligibility threshold induces strategic behavioral responses. For this analysis, we use the unaggregated transaction-level sample and restrict it to the EUR 120 to EUR 180 range around the threshold. Following the logic of a bunching analysis (Burgstahler and Dichev 1997; Saez 2010; Chetty et al. 2011; Best and Kleven 2018; Coles et al. 2022), we analyze the distribution of declared transaction values around the EUR 150 cutoff, separately for B2C and B2B transactions and for pre- and post-reform periods. An accumulation of B2C transactions just below the threshold in the post-reform period, accompanied by a corresponding decline just above it, would indicate that firms adjust their behavior to qualify for the simplified portal. Since B2B transactions are not eligible for the IOSS, their distribution serves as a benchmark against which to assess any irregularities in the B2C distribution. To quantify this pattern, we complement the visual analysis with a DiD regression approach following Bernard et al. (2018), which estimates whether B2C transactions became significantly more likely than B2B transactions to be declared below the EUR 150 threshold after the reform.

Third, we examine the mechanisms underlying the response around the threshold to distinguish between real and reporting responses. We first analyze the evolution of declared transaction values around the threshold, in a DiD design using B2B transactions as a control group to assess whether average declared prices decline after the reform. Since a decline in average transaction values is consistent with both genuine price reductions and downward misreporting, we additionally plot average transaction weights in narrow price bins around the EUR 150 threshold. A decline in transaction weight within a given price bin after the reform, with no comparable decline for B2B transactions would be consistent with parcel splitting, whereby firms divide shipments into multiple parcels each assigned a lower declared value to remain below the threshold without any change in the goods actually sold.

As in the OSS analysis, our estimates capture the intent-to-treat effect of the reform. While not all eligible B2C imports are processed through the IOSS portal, adoption rates are substantial. Based on our data, the average take-up rate over the post-reform period is 73%, increasing from 57% in 2022 to 81% in 2023, suggesting growing familiarity with the system over time. The high and increasing adoption rate suggests that the estimated effects largely reflect behavioral changes induced by the introduction of the IOSS.

4.2 Data, Sample and Descriptives

Our analysis of the IOSS draws on a third administrative data source: transaction-level customs data on imports into Germany from non-EU countries, provided by the German customs authority for the years 2018 to 2023.²⁶ The dataset covers all imports from non-EU countries into Germany with declared transaction values up to EUR 1,500, and contains information on the declared transaction value and weight, year, product type, country of origin, and transaction type (B2C or B2B).

Appendix Table B.2 reports our sample selection procedure. We exclude all intra-EU transactions as well as private transactions, transactions with missing or invalid country codes, and transactions involving goods subject to excise duties or originating from Great Britain. We restrict the sample to transactions valued above EUR 30 to avoid confounding effects from the simultaneous abolition of the EUR 22 VAT exemption, and drop all transactions from the transitional year 2021, as the reform was implemented in July of that year and the available data contain only a yearly indicator, making it impossible to determine whether a given transaction occurred before or after the reform date.

Classifying transactions as B2C or B2B presents an additional challenge. While the dataset contains a transaction type indicator, this indicator is recorded only for the post-reform period and is unavailable for pre-reform observations. We therefore implement a data-driven imputation strategy. For each four-digit product code, we calculate the relative frequency of B2C and B2B designations among classified post-reform transactions and assign unclassified pre-reform transactions of that product type to whichever category predominates. This approach may result in a slightly higher share of B2C transactions in the pre-period, which would bias our estimates downward and thus represents a conservative assumption.²⁷ After applying all restrictions, our final sample comprises approximately 58 million transactions.

Our analyses draw on two subsamples derived from this transaction-level dataset. For the DiD analysis, we aggregate transactions valued between EUR 30 and EUR 150 to the product-country-year level, the most granular aggregation available given the absence of firm identifiers. For the distributional and mechanism analyses, we use the unaggregated

²⁶The dataset is sourced from the ATLAS import registration system and from ATLAS Impost, a registration system introduced alongside the VAT e-commerce package to handle transactions valued up to EUR 150.

²⁷To validate that our classification approach does not drive our results, we additionally implement two alternative B2C classifications. The first is a strict product-based classification that assigns each transaction to the same B2C or B2B category in both the pre- and post-reform periods. The second is a more lenient manual classification based on a broader set of product codes. The results under both alternative approaches are reported in the Appendix and are qualitatively consistent with the baseline findings.

transaction-level sample restricted to the EUR 120 to EUR 180 range around the threshold. Where necessary, we adjust the window around the threshold for specific analyses as described below.

Table 6 presents pre-reform summary statistics for both subsamples, covering the years 2018 to 2020 and reported separately for B2C and B2B transactions. Panel A reports statistics at the product-country-year level, where the average B2C cell records 232 transactions and a total sales volume of EUR 15,780 per product-country-year. Panel B reports statistics at the transaction level. For B2C transactions between EUR 120 and EUR 180, the average transaction value is EUR 147, with a median weight of 4 kg. B2B transactions are very comparable in terms of their transaction value but are heavier than their B2C counterparts.²⁸

4.3 Results

We start by testing whether the simplification effect documented in the OSS setting extends to the IOSS context, where access to the simplified portal is restricted to transactions valued below EUR 150. Table 7 reports DiD estimates comparing B2C and B2B import transactions valued below the EUR 150 eligibility threshold before and after the introduction of the IOSS. Consistent with the OSS results in the previous section, we find a large, positive, and statistically significant increase in B2C activity following the reform, both in the number of transactions and in sales volume, indicating that the simplification effect extends to non-EU firms exporting to the EU. Figure 4 confirms the parallel trends assumption, with pre-reform coefficients for 2018 to 2020 statistically indistinguishable from zero. As in the OSS setting, the increase in transaction counts exceeds the increase in sales volume. Notably, however, both effects are considerably larger in magnitude than those documented for the OSS, suggesting that the eligibility threshold may be generating additional behavioral responses beyond the baseline simplification effect.

We therefore turn to the second step and examine the distribution of declared transaction values around the threshold to assess whether the pronounced response reflects real increases in business activity, strategic reporting behavior, or both. We plot reported transaction values around the EUR 150 threshold for B2C and B2B imports, separately for the pre- and post-reform periods. Figure 5 displays histograms of transaction frequencies in

²⁸Additional descriptive statistics are provided in Appendix Tables B.3 and B.4, which report the main countries of origin for imports into Germany and the most frequently imported product categories among B2C transactions.

the EUR 120 to EUR 180 range. Panels (a) and (b) show the distribution for B2C imports and Panels (c) and (d) for B2B imports.

In the pre-reform period, we observe a modest concentration of transactions just below the EUR 150 threshold across both panels, consistent with the pre-existing tariff exemption for goods valued below this cutoff. More importantly, Panel (b) reveals a pronounced shift in the post-reform B2C distribution. The number of transactions just below the threshold increases substantially relative to the pre-reform period, while a corresponding decline is visible just above it, producing a clear hole in the distribution. This pattern is not mirrored in the B2B distributions shown in Panels (c) and (d), where the distribution remains smooth around the threshold both before and after the reform, consistent with B2B transactions being ineligible for the IOSS. The divergence between the B2C and B2B distributions in the post-reform period suggests that the simplified reporting regime has incentivized firms to declare a greater share of transactions below the EUR 150 eligibility threshold.²⁹

We conduct several tests to rule out alternative explanations for this pattern. First, we verify that the results are not driven by our B2C classification approach by replicating the analysis using both a strict product-based classification, which assigns each transaction to the same category in the pre- and post-reform periods, and a more lenient B2C definition. Appendix Figures B.2 and B.3 confirm qualitatively similar patterns under both approaches.

Second, we address the concern that the observed leftward shift in declared transaction values may reflect tariff evasion rather than a strategic response to the IOSS. To assess this, we separately plot the transaction value distribution for B2C transactions that are subject to tariffs and those that are not. Figure 6 shows that the leftward shift appears in both groups. Notably, for goods not subject to any tariff, the only remaining financial incentive to declare transactions below EUR 150 is access to the simplified VAT reporting portal, which directly supports the interpretation that the IOSS is the primary driver of the observed pattern.

Third, we examine whether the distributional shift is consistent across source countries. Figure 7 presents transaction value distributions for Germany's two most important non-EU trading partners, China and the United States, respectively. Both countries exhibit

²⁹Since the pre-reform period spans three years and the post-reform period two years, the absolute number of transactions is not directly comparable across periods. The analysis therefore focuses on the shape of the distribution rather than aggregate transaction counts. This interpretation is further supported by Appendix Figure B.1, which plots relative transaction frequencies.

a clear leftward shift in the B2C distribution in the post-reform period, though the effect is somewhat more pronounced for China.³⁰

To quantify the distributional shift, we follow Bernard et al. (2018) and estimate the likelihood that a transaction is declared below the EUR 150 threshold using a DiD framework. The unit of observation is a single transaction. The dependent variable is an indicator equal to one if the declared transaction value falls below EUR 150 and zero otherwise, and the coefficient of interest captures whether B2C transactions became more likely than B2B transactions to be reported below the cutoff after the reform. We estimate the specification using logit with product fixed effects (Column 1) and country fixed effects (Column 2), and using PPML with product-country fixed effects (Column 3), across three bandwidths of EUR ± 1 , ± 10 , and ± 30 around the threshold.

Table 8 presents the results. Across all specifications and bandwidths, the coefficient on the B2C and post-reform interaction is positive and statistically significant. In the narrowest bandwidth of EUR ± 1 around the threshold, the logit estimate implies that B2C transactions became approximately 34% more likely to be declared below the threshold relative to B2B transactions following the reform. Expanding the bandwidth to EUR ± 30 yields a larger estimate of approximately 49%, suggesting that the shift in declared values extends across a broader range of transaction values rather than being confined to the immediate vicinity of the threshold. The results are robust across all bandwidths and fixed effect structures, consistent with the graphical evidence in Figure 5.

4.4 Mechanisms

Our findings show that firms shift transactions from above to below the threshold. We next examine the mechanisms underlying this distributional shift. As outlined above, firms may respond along the real margin by genuinely reducing prices or expanding the supply of lower-priced goods, or along the reporting margin by misreporting the value of transactions without any change in underlying business activity. We assess the relevance of each of these channels.

The distributional shift is consistent with a decline in average declared transaction values, which we first quantify using a DiD framework. We estimate a DiD regression comparing mean and median transaction values for B2C and B2B imports within the EUR 120 to EUR 180 range before and after the implementation of the IOSS at the product-country-

³⁰The corresponding B2B distributions are reported in Appendix Figure B.4. We also replicate the analysis for Switzerland, Japan, and Norway and find qualitatively consistent results across all three countries.

year level. Table 9 Panel A reports the results. Across all specifications, B2C transaction values decline significantly relative to B2B transactions, with Column (1) implying an average reduction of EUR 2.31 within the estimation window. The event-study results in Panels (a) and (b) of Figure 8 support this decline and confirm flat pre-trends, with pre-reform coefficients for 2018 to 2020 statistically indistinguishable from zero. This decline is consistent with price reductions along the real margin, but does not rule out downward misreporting of transaction values. Importantly, a pure real response cannot fully rationalize the distributional evidence, since marginal price reductions could generate excess mass just below the threshold but cannot explain the pronounced missing mass extending well beyond it.

To assess whether reporting behavior is additionally present, we examine whether transaction weights decline following the reform. One way to achieve a lower transaction value is for firms to split transactions that are truly priced above 150 EUR into multiple lighter parcels, each reported separately below the threshold. If parcel splitting is occurring, average transaction weights should decline for B2C imports around the threshold following the reform, as a given shipment is divided into smaller units rather than dispatched as a single transaction. This is consistent with the strategic reporting margin, as firms only split existing parcels into several parcels without adjusting any underlying business activities.

To shed light on potential parcel splitting, we estimate the same DiD specification using mean and median transaction weights at the product-country-year level.³¹ Panel B of Table 9 reports the results. Across all specifications, the effect is negative and statistically significant, with Column (1) implying a reduction of approximately 0.82 kg in the average weight of B2C transactions relative to comparable B2B transactions following the reform. Panels (c) and (d) of Figure 8 confirm flat pre-trends and a statistically significant post-reform decline in B2C transaction weights. This decline is consistent with firms splitting higher-valued transactions into multiple lighter parcels, each assigned a lower declared value, to qualify for the simplified IOSS portal.³²

³¹To limit the influence of outliers, we exclude observations above the 75th percentile of the pre-reform weight distribution separately for B2C and B2B transactions, corresponding to cutoffs of 25 kg and 95 kg, respectively. Reported transaction weights exhibit substantial dispersion and may be affected by data errors, such as confusion between grams and kilograms. We therefore treat transactions weighing several tons but valued below EUR 150 as implausible. Because B2B transactions are considerably heavier on average, as shown in Table 6, we apply group specific thresholds.

³²Appendix Figure B.6 provides complementary descriptive support for this result. Plotting mean transaction weights for each EUR 1 bin of reported values in the EUR 120 to EUR 180 range, shows that average B2C transaction weights decline across the entire value range after the reform, whereas B2B weights remain

We acknowledge that the weight trimming required to address outliers may limit the precision of this estimate. To corroborate the finding using an approach that does not require trimming, we return to the transaction-level data and examine whether the distributional shift in reported values is more pronounced for goods that can plausibly be divided into multiple parcels. We manually classify all four-digit product codes into divisible goods, such as clothing and footwear, and non-divisible goods, such as large electronics or furniture, and replicate the histogram analysis separately for each group. Figure 9 shows that the leftward shift in the B2C distribution around the EUR 150 threshold is substantially more pronounced for divisible goods, consistent with the DiD evidence and providing further support for strategic parcel splitting as an underlying mechanism. Taken together, the transaction value and weight evidence indicates that both real and reporting responses are present. Firms adjust their pricing behavior at the margin to bring transactions below the threshold, while simultaneously splitting higher-valued shipments into multiple parcels to misreport their declared values, representing an unintended consequence of the threshold-based design of the IOSS.

5 Conclusion

Simplifying regulation and reducing bureaucratic costs for businesses are stated goals of economic policy worldwide, yet empirical evidence on the economic effects of simplification and on firms' behavioral responses remains limited. We study the EU's 2021 VAT reform, which introduced the OSS and IOSS portals to substantially simplify cross-border VAT registration and reporting for B2C transactions, using novel high-frequency transaction-level administrative data from Germany.

Our analysis identifies two complementary sets of firm responses to the reform. For the OSS portal, which simplified intra-EU B2C exports for German firms without imposing an eligibility threshold, we find that treated firms substantially increased their cross-border business activity. Transaction counts rose by roughly 34%, accompanied by greater product diversification, and expansion into new destination markets. Mechanism tests are consistent with a reduction in both variable and fixed compliance costs following the reform. Firms appear to pass at least part of these savings on to consumers in the form of lower prices, as evidenced by a decline in mean product prices. Firms already selling to consumers in multiple countries respond more strongly by expanding their product

stable.

scope, consistent with freed-up variable compliance resources being used to deepen existing trading activities. The reform also appears to lower the fixed costs of entering new destination markets. While the geographic expansion effect is similar for firms with different pre-reform geographic scope, it is concentrated among small and medium-sized firms, for whom destination-specific VAT registration and filing requirements were likely most binding.

For the IOSS portal, which restricts access to the simplified regime to transactions valued below EUR 150, the reform generated both real effects and strategic reporting responses. On the real side, transaction counts and sales volumes for eligible B2C imports increased substantially, mirroring the OSS findings. The post-reform distribution of transactions exhibits pronounced excess mass just below the threshold and a corresponding missing mass above it, a pattern that real responses alone cannot fully rationalize. Examining the underlying mechanisms, we find that average declared transaction values decline following the reform, consistent with both genuine price reductions and downward misreporting, while transaction weights decline within every price bin, evidence that points specifically to firms splitting higher-valued parcels into multiple lighter parcels to fall below the threshold. Together our results reflect both real activity increases and strategic reporting behavior, the latter constituting an unintended consequence of pairing simplification with an eligibility threshold.

Overall, these findings speak to a broader tension in regulatory simplification. While reducing compliance costs can meaningfully stimulate economic activity, eligibility thresholds, a common design feature of simplified tax regimes, can simultaneously induce strategic reporting behavior that undermines the policy's objectives. The co-existence of real and reporting responses within a single setting illustrates precisely the interaction that prior literature has called for researchers to examine (Lester and Olbert 2025), and demonstrates that the behavioral consequences of simplification depend critically on how the simplified regime is designed. Our findings are directly relevant to ongoing policy debates, including EU proposals to extend the IOSS to all imports regardless of value, and contribute to a broader understanding of how the design of simplified tax regimes shapes firm behavior and the extent to which simplification can deliver on its stated economic objectives.

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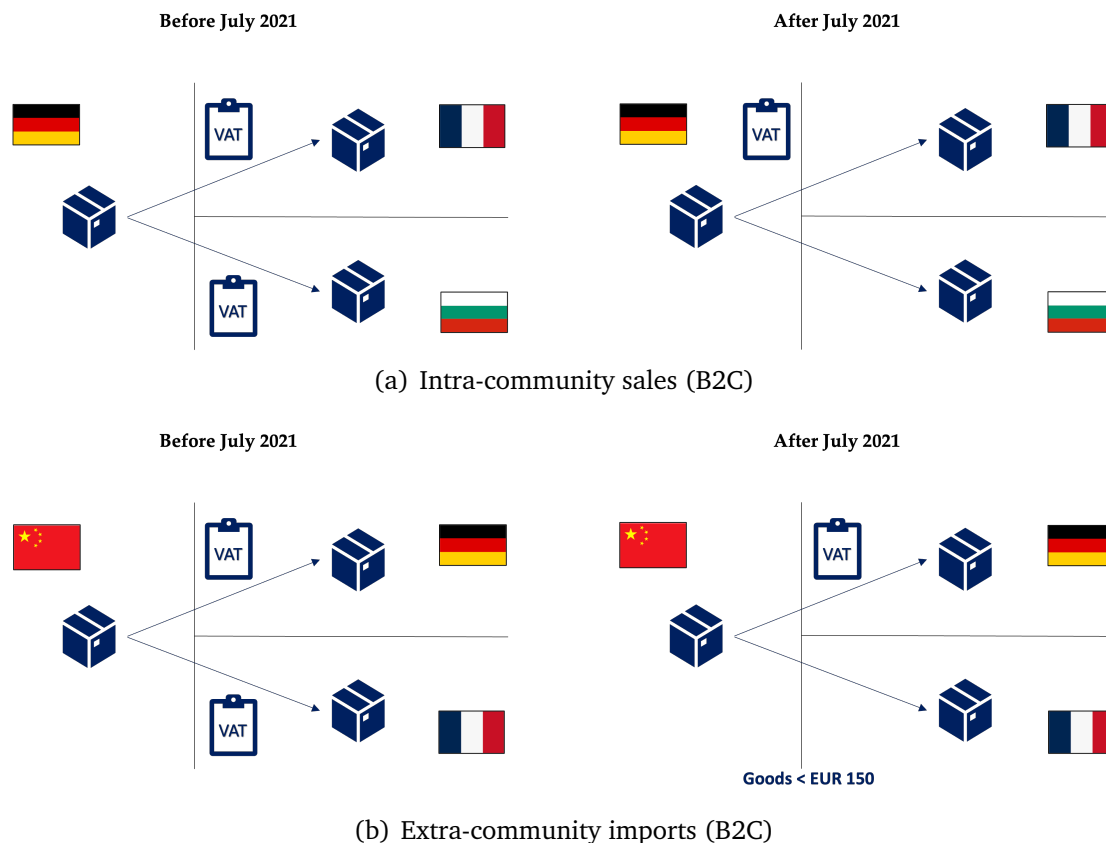
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Tables and Figures

Figure 1: Simplifying VAT Reporting



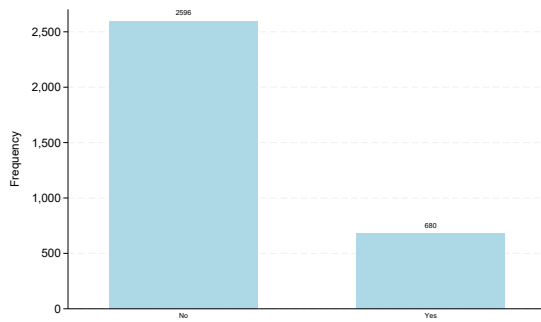
Notes: This figure illustrates the regulatory changes introduced by the EU VAT e-commerce package of July 2021. Panel (a) summarizes the changes for intra-community B2C sales under the One Stop Shop (OSS), which replaced the requirement for EU firms to register and file VAT separately in each destination member state with a single registration and unified reporting portal in the firm's home member state. Panel (b) summarizes the changes for extra-community B2C imports under the Import One Stop Shop (IOSS), which introduced an analogous single-portal system for non-EU firms selling goods to EU consumers, applicable to transactions valued at EUR 150 or less. In both panels, the left side depicts the pre-reform registration and reporting environment and the right side depicts the simplified post-reform system.

Table 1: Key Regulatory Changes under the EU VAT E-Commerce Package

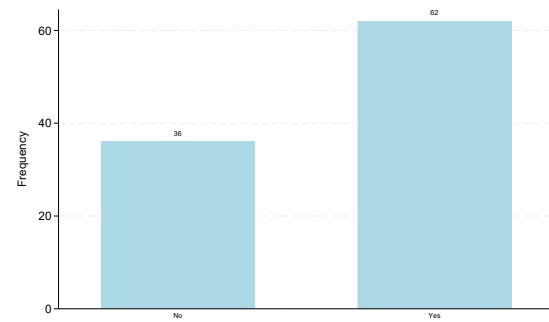
Panel A: Intra-community sales (B2C)			
		Before July 2021	After July 2021
VAT registration & reporting		In each destination country	Via OSS in one country
VAT rate		Of the destination country	Of the destination country
Panel B: Extra-community imports (B2C)			
		Before July 2021	After July 2021
VAT registration & reporting	EUR 0 – 150	In each destination country	Via IOSS in one country
	> EUR 150	In each destination country	In each destination country
(Import) VAT rate		Of the destination country	Of the destination country
(Import) VAT	EUR 0 – 22	Import VAT exempt	VAT via IOSS
	EUR 22 – 150	Import VAT	VAT via IOSS
	> EUR 150	Import VAT	Import VAT
Tariffs	EUR 0 – 150	Tariff exempt	Tariff exempt
	> EUR 150	Tariff applicable for selected products	Tariff applicable for selected products

Notes: This table summarizes the key regulatory changes introduced by the EU VAT e-commerce package in July 2021. Panel A outlines the changes for intra-community B2C sales by EU firms to consumers in other EU member states. Prior to the reform, such firms were required to register for VAT and file returns separately in each destination country. The One Stop Shop (OSS), introduced in July 2021, replaced this fragmented system with a single registration portal in the firm's home member state, through which all intra-EU B2C sales can be reported and VAT remitted. The applicable VAT rate remains that of the destination country in both regimes. Panel B outlines the changes for extra-community B2C imports by non-EU firms selling goods to EU consumers, categorized by transaction value. The Import One Stop Shop (IOSS) introduced an analogous single-portal system for these firms, but applies only to transactions valued at EUR 150 or less. For transactions above this threshold, the pre-reform requirement to register and report separately in each destination country remains unchanged. The reform also abolished the pre-existing EUR 22 import VAT exemption for low-value goods, meaning that all imports became subject to VAT regardless of value. The tariff exemption for goods valued below EUR 150, which predates the reform, remains unchanged. Tariffs apply only to selected product categories. Goods not subject to tariffs remain tariff-free regardless of their transaction value.

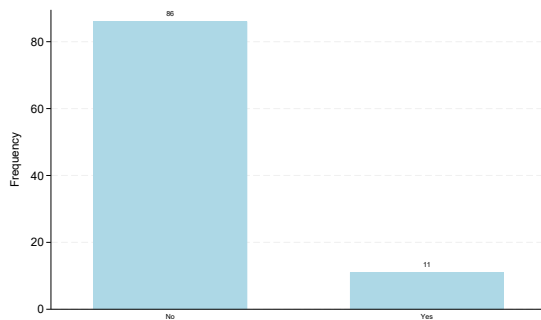
Figure 2: Survey Evidence on Perceptions of the Simplification Reform



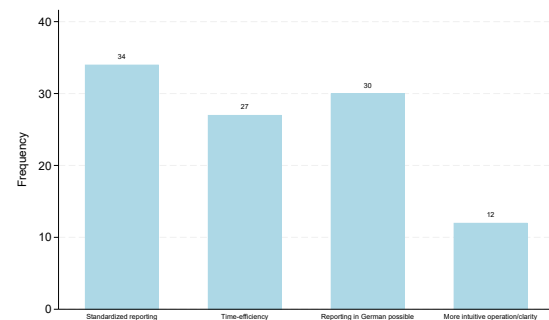
(a) Has your company sold goods/services to final customers in other EU member states in recent years (2020 to date)? $N=3,276$



(b) Has the VAT declaration of B2C transactions to other EU member states been simplified by using the OSS portal in your company? $N=99$



(c) Does your company deliver to additional EU member states as a result of using the OSS portal? $N=99$



(d) To what extent has the OSS portal simplified the VAT declaration of B2C transactions? $N=62$

Notes: This figure shows the main results of the survey administered by the German Business Panel in spring 2025 with a total of 3,276 respondents, of which 99 actively use the OSS portal.

Table 2: OSS – Descriptive Statistics

Panel A:		B2C firms				
	<i>Mean</i>	<i>SD</i>	<i>P1</i>	<i>P50</i>	<i>P99</i>	<i>Obs</i>
<i>Dependent variables:</i>						
Count	12.46	299.90	1.00	1.00	159	2,164,005
Sales (EUR)	71,730.34	2,035,667.45	19.00	2,471.00	768,542	2,164,005
Goods	1.45	3.21	1.00	1.00	15	2,164,005
Countries	1.69	2.88	1.00	1.00	18	1,942,829
<i>Control variables:</i>						
Pre mean no. employees	40.46	596.42	0.00	2.00	515	1,910,398
Pre mean total sales (mio. EUR)	10.83	190.46	0.03	0.72	106.76	1,910,398
Pre mean count (DE-EU)	11.40	230.26	1.00	1.00	145	1,929,071
Pre mean EU-sales (mio. EUR)	0.07	1.85	0.00	0.00	0.67	1,929,071
<i>No. of firms</i>						111,081
Panel B:		B2B firms				
	<i>Mean</i>	<i>SD</i>	<i>P1</i>	<i>P50</i>	<i>P99</i>	<i>Obs</i>
<i>Dependent variables:</i>						
Count	21.12	190.82	1.00	1.00	352	6,464,674
Sales (EUR)	358,847.14	7,386,305.71	43.00	10,715.00	5,205,454	6,464,674
Goods	2.06	3.73	1.00	1.00	20	6,464,674
Countries	3.33	4.81	1.00	1.00	22	5,929,356
<i>Control variables:</i>						
Pre mean no. employees	63.94	975.61	0.00	8.86	739	6,043,055
Pre mean total sales (mio. EUR)	26.48	641.28	0.04	2.48	292.99	6,043,055
Pre mean count (DE-EU)	21.39	192.24	1.00	1.00	345	6,121,399
Pre mean EU-sales (mio. EUR)	0.35	6.79	0.00	0.02	4.75	6,121,399
<i>No. of firms</i>						236,923

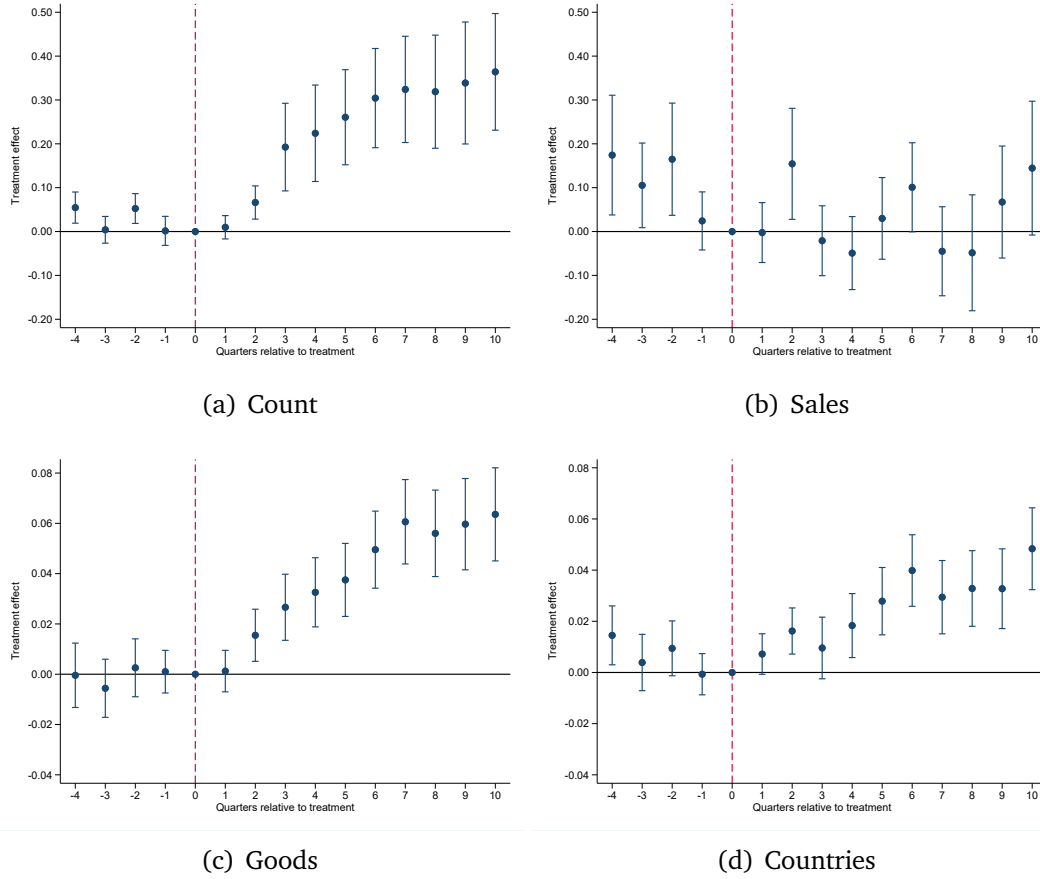
Notes: This table reports the descriptive statistics for the main variables used in the OSS analysis, covering the period January 2018 to December 2023. The data combine the German foreign trade statistics with VAT return data and are aggregated at the firm-month level, such that each observation represents a firm's total cross-border activity in a given month. Panel A reports statistics for treated B2C firms and Panel B for control B2B firms. Appendix A.1 provides variable definitions.

Table 3: OSS – Effect of Simplified Reporting on Business Activity

Panel A: Fixed effects				
	<i>Count</i> (1)	<i>Sales</i> (2)	<i>Goods</i> (3)	<i>Countries</i> (4)
B2C x Post	0.298*** (0.062)	0.072 (0.045)	0.039*** (0.006)	0.010*** (0.005)
<i>Firm FE</i>	X	X	X	X
<i>Year-month FE</i>	X	X	X	X
<i>Pseudo R2</i>	0.94	0.95	0.48	0.57
<i>Observations</i>	8,628,679	8,628,679	8,628,679	7,864,644
Panel B: Fixed effects and cross-border trade controls				
	<i>Count</i> (1)	<i>Sales</i> (2)	<i>Goods</i> (3)	<i>Countries</i> (4)
B2C x Post	0.295*** (0.070)	0.081 (0.052)	0.053*** (0.006)	0.020*** (0.005)
<i>Firm FE</i>	X	X	X	X
<i>Year-month FE</i>	X	X	X	X
<i>Pseudo R2</i>	0.94	0.95	0.49	0.58
<i>Observations</i>	8,050,470	8,050,470	8,050,470	7,286,435
Panel C: Fixed effects and firm-level controls				
	<i>Count</i> (1)	<i>Sales</i> (2)	<i>Goods</i> (3)	<i>Countries</i> (4)
B2C x Post	0.285*** (0.058)	0.066 (0.043)	0.047*** (0.006)	0.022*** (0.005)
<i>Firm FE</i>	X	X	X	X
<i>Year-month FE</i>	X	X	X	X
<i>Pseudo R2</i>	0.94	0.95	0.49	0.58
<i>Observations</i>	7,953,453	7,953,453	7,953,453	7,190,817

Notes: This table presents DiD estimates of the effect of the OSS on cross-border business activity, estimated using PPML. The dependent variables are measured at the firm-month level and capture four dimensions of a firm's cross-border activity: the number of transactions (Column 1), total sales volume (Column 2), the number of goods traded (Column 3) and the number of destination countries (Column 4). The treatment group consists of B2C firms and the control group of B2B firms. The post-reform period covers July 2021 to December 2023 and the pre-reform period January 2018 to June 2021. All specifications include firm fixed effects and year-month fixed effects. Panel B additionally includes pre-reform cross-border trade controls interacted with a linear time trend, and Panel C replaces these with firm-level pre-reform controls. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Figure 3: OSS – Dynamic Effect of Simplified Reporting on Business Activity



Notes: This figure presents event-study estimates of the effect of the OSS introduction on cross-border business activity, obtained by replacing the *Post* indicator in Equation 1 with separate year quarter indicators. To display the full post-reform period, the monthly firm-level panel is aggregated to the firm quarter level. The treatment group consists of German B2C firms and the control group of B2B firms. The specification includes firm and year quarter fixed effects and pre-reform cross-border trade controls interacted with a linear time trend. Coefficients are plotted relative to the quarter immediately preceding the reform (Q2 2021), marked by the dashed vertical line. Standard errors are clustered at the firm level and the blue bands denote 95% confidence intervals.

Table 4: OSS – Effect of Simplified Reporting on Business Activity
(Treatment Intensity)

Panel A	Fixed effects			
	Count (1)	Sales (2)	Goods (3)	Countries (4)
Intra-EU B2C x Post	0.196** (0.088)	0.044 (0.068)	0.098*** (0.012)	0.170*** (0.010)
Firm FE	X	X	X	X
Year-month FE	X	X	X	X
Pseudo R2	0.95	0.94	0.49	0.48
Observations	1,547,105	1,547,105	1,547,105	1,530,582
Panel B:	Fixed effects and cross-border trade controls			
	Count (1)	Sales (2)	Goods (3)	Countries (4)
Intra-EU B2C x Post	0.149** (0.063)	0.016 (0.071)	0.113*** (0.013)	0.191*** (0.013)
Firm FE	X	X	X	X
Year-month FE	X	X	X	X
Pseudo R2	0.95	0.94	0.49	0.48
Observations	1,547,105	1,547,105	1,547,105	1,530,582

Notes: This table presents DiD estimates of the effect of the OSS on cross-border business activity using an alternative treatment intensity specification, estimated using PPML. This specification restricts the sample to B2C firms only. The treatment group consists of B2C firms with more than 60% of pre-reform exports directed to EU member states, while the control group consists of B2C firms with less than 40% of pre-reform exports directed to EU member states. The dependent variables are measured at the firm-month level and capture four dimensions of a firm's cross-border activity: the number of transactions (Column 1), total sales volume (Column 2), the number of goods traded (Column 3) and the number of destination countries (Column 4). The post-reform period covers July 2021 to December 2023, and the pre-reform period covers January 2018 to June 2021. All specifications include firm fixed effects and year-month fixed effects. Panel B additionally includes pre-reform cross-border trade controls interacted with a linear time trend. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Table 5: OSS: Mechanism Tests

Panel A:		Price pass-through			
	<i>Price</i> (1)	<i>Weight</i> (2)	<i>Count</i> (3)	<i>Goods</i> (4)	<i>Countries</i> (5)
B2C x Post	-0.026** (0.011)	0.002 (0.012)			
<i>Pseudo R2</i>	0.82	0.91			
<i>Observations</i>	26,673,636	26,673,636			
Panel B:		Pre-reform countries			
	<i>Price</i> (1)	<i>Weight</i> (2)	<i>Count</i> (3)	<i>Goods</i> (4)	<i>Countries</i> (5)
Single: B2C x Post			0.442*** (0.104)	0.023*** (0.005)	0.019*** (0.005)
<i>Pseudo R2</i>			0.45	0.03	0.02
<i>Observations</i>			4,513,956	4,513,956	4,513,947
Multiple: B2C x Post			0.277*** (0.073)	0.081*** (0.011)	0.020*** (0.009)
<i>Pseudo R2</i>			0.94	0.58	0.51
<i>Observations</i>			2,749,974	2,749,974	2,749,837
Panel C:		Firm size			
	<i>Price</i> (1)	<i>Weight</i> (2)	<i>Count</i> (3)	<i>Goods</i> (4)	<i>Countries</i> (5)
Small: B2C x Post			0.245*** (0.036)	0.041*** (0.007)	0.022*** (0.006)
<i>Pseudo R2</i>			0.84	0.30	0.46
<i>Observations</i>			6,692,272	6,692,272	5,961,516
Medium: B2C x Post			0.280*** (0.077)	0.136*** (0.024)	0.054*** (0.016)
<i>Pseudo R2</i>			0.91	0.58	0.54
<i>Observations</i>			982,376	982,376	952,972
Large: B2C x Post			0.333** (0.130)	0.071*** (0.023)	-0.006 (0.022)
<i>Pseudo R2</i>			0.95	0.69	0.56
<i>Observations</i>			278,805	278,805	276,329

Notes: This table presents mechanism tests for the effect of the OSS introduction on cross-border business activity using OLS or PPML. Panel A examines the price pass-through channel and is estimated using OLS, with the logarithm of mean transaction value and the logarithm of mean transaction weight as outcome variables at the firm-product-month level. Panels B and C are estimated using PPML. Panel B splits the sample by the number of destination countries served prior to the reform, while Panel C splits the sample by firm size, defined as small (<50 employees), medium (50–250 employees), and large (>250 employees). The treatment group consists of B2C firms and the control group of B2B firms. The post-reform period covers July 2021 to December 2023 and the pre-reform period January 2018 to June 2021. All specifications include firm fixed effects, year-month fixed effects, and cross-border trade controls. Standard errors are clustered at the firm level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Table 6: IOSS – Descriptive Statistics

Panel A:		Product-Country Year Sample				
		B2C Transactions between EUR 30 and EUR 150				
	Mean	SD	P1	P50	P99	Obs
Count	231.92	3,139.48	1.00	4.00	3,184.50	43,600
Sales (EUR)	15,781.48	185,102.60	31.00	332.00	235,472.00	43,600
Log (Pre sales)	5.32	3.14	0.00	5.59	12.21	43,600
		B2B Transactions between EUR 30 and EUR 150				
	Mean	SD	P1	P50	P99	Obs
Count	120.31	1,797.71	1.00	3.00	1,908.00	36,163
Sales (EUR)	9,755.06	158,137.29	31.00	261.00	156,413.00	36,163
Log (Pre sales)	5.15	3.09	0.00	5.44	11.97	36,163
Panel B:		Transaction Year Sample				
		B2C Transactions between EUR 120 and EUR 180				
	Mean	SD	P1	P50	P99	Obs
Transaction value (EUR)	146.97	17.53	120.00	145.00	180.00	1,695,927
Weight (kg)	460.67	11,965.47	0.10	4.00	11,176.40	1,695,927
		B2B Transactions between EUR 120 and EUR 180				
	Mean	SD	P1	P50	P99	Obs
Transaction value (EUR)	146.65	17.32	120.00	144.00	180.00	1,384,802
Weight (kg)	707.75	107,388.36	0.10	5.00	13,319.70	1,384,802

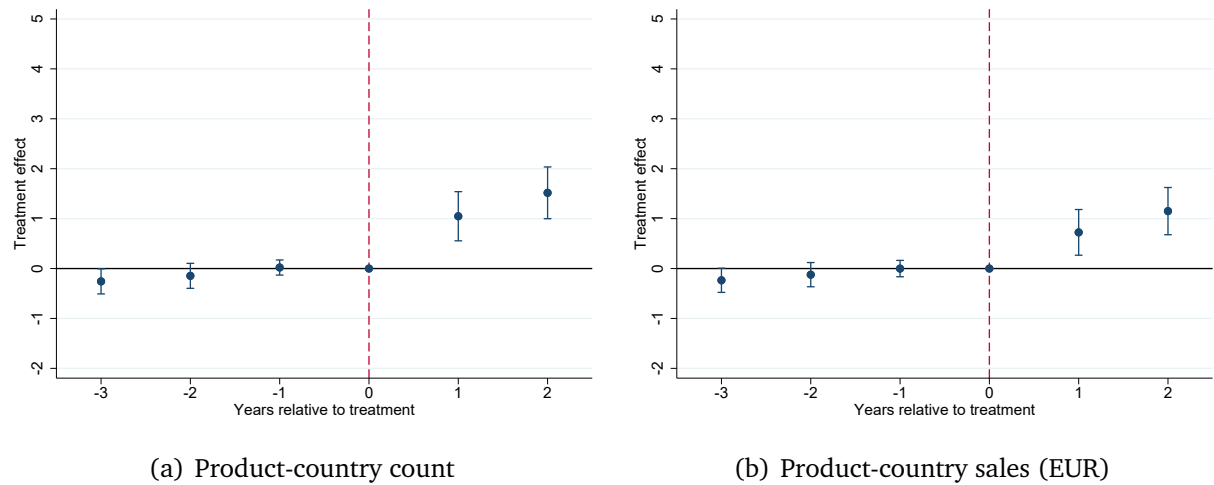
Notes: This table reports pre-reform descriptive statistics for the main variables used in the IOSS analysis, covering the period 2018 to 2020. The data are sourced from the German customs authority. Panel A reports descriptive statistics at the product-country year level for the aggregated sample used in Table 7, separately for B2C and B2B transactions valued between EUR 30 and EUR 150. Panel B reports descriptive statistics at the transaction year level, where each observation represents a single import transaction into Germany from a non-EU country. Statistics are reported separately for B2C and B2B transactions, each group between EUR 120 and EUR 180. Appendix B.1 provides variable definitions.

Table 7: IOSS – Effect of Simplified Reporting on Below-threshold Business Activity

	EUR 30 to EUR 150			
	<i>Product-country count</i>		<i>Product-country sales</i>	
	(1)	(2)	(3)	(4)
B2C × Post	1.369*** (0.269)	1.363*** (0.269)	1.047*** (0.239)	1.039*** (0.240)
<i>Product FE</i>	X		X	
<i>Product × Country FE</i>		X		X
<i>Year FE</i>	X	X	X	X
<i>Controls</i>	X	X	X	X
<i>Pseudo R²</i>	0.837	0.908	0.851	0.912
<i>Obs</i>	178,135	178,135	160,992	160,992

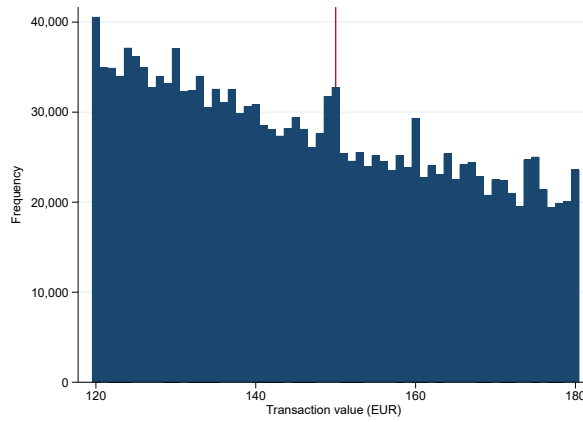
Notes: This table presents DiD estimates of the effect of the IOSS introduction on import activity for transactions valued between EUR 30 and EUR 150, estimated using PPML. The analysis is conducted at the product-country-year level. The dependent variables are the number of transactions (*Count*) and total sales volume (*Sales*), aggregated at the product-country-year level. The treatment group consists of B2C imports and the control group of B2B imports. Columns (1) and (3) include product and year fixed effects, while Columns (2) and (4) include product-country and year fixed effects. All specifications include controls for the pre-treatment level of sales. The post-reform period covers 2022 and 2023 and the pre-reform period 2018 to 2020. Standard errors are clustered at the product level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Figure 4: IOSS – Dynamic Effect of Simplified Reporting on Below-threshold Business Activity

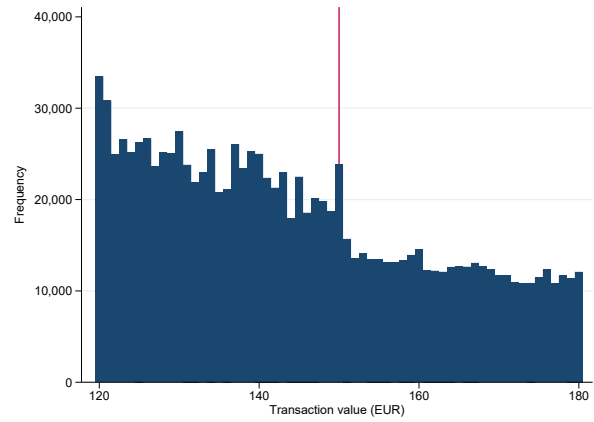


Notes: This figure presents event-study estimates of the effect of the IOSS introduction on import activity for transactions valued between EUR 30 and EUR 150 estimated using PPML, obtained by replacing the *Post* indicator with separate year indicators. Panels (a) show the effect on the count of transactions and Panel (b) shows the effect on sales. All dependent variables are aggregated at the product-country-year level and the sample is restricted to transactions valued between EUR 30 and EUR 150. All specifications include product-country and year fixed effects, as well as controls for the pre-treatment level of sales. Coefficients are plotted relative to the year of the reform (2021), marked by the dashed vertical line. Standard errors are clustered at the product level and the shaded bands denote 95% confidence intervals.

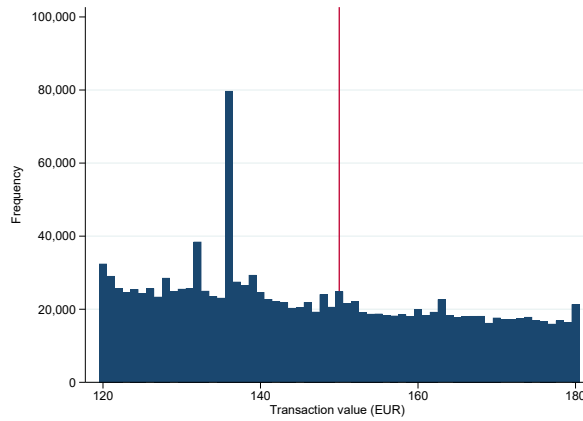
Figure 5: IOSS – Distribution of B2C and B2B Transaction Values



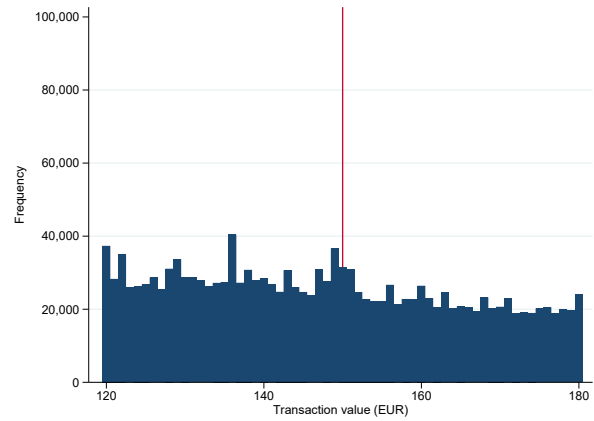
(a) PRE: B2C imports



(b) POST: B2C imports



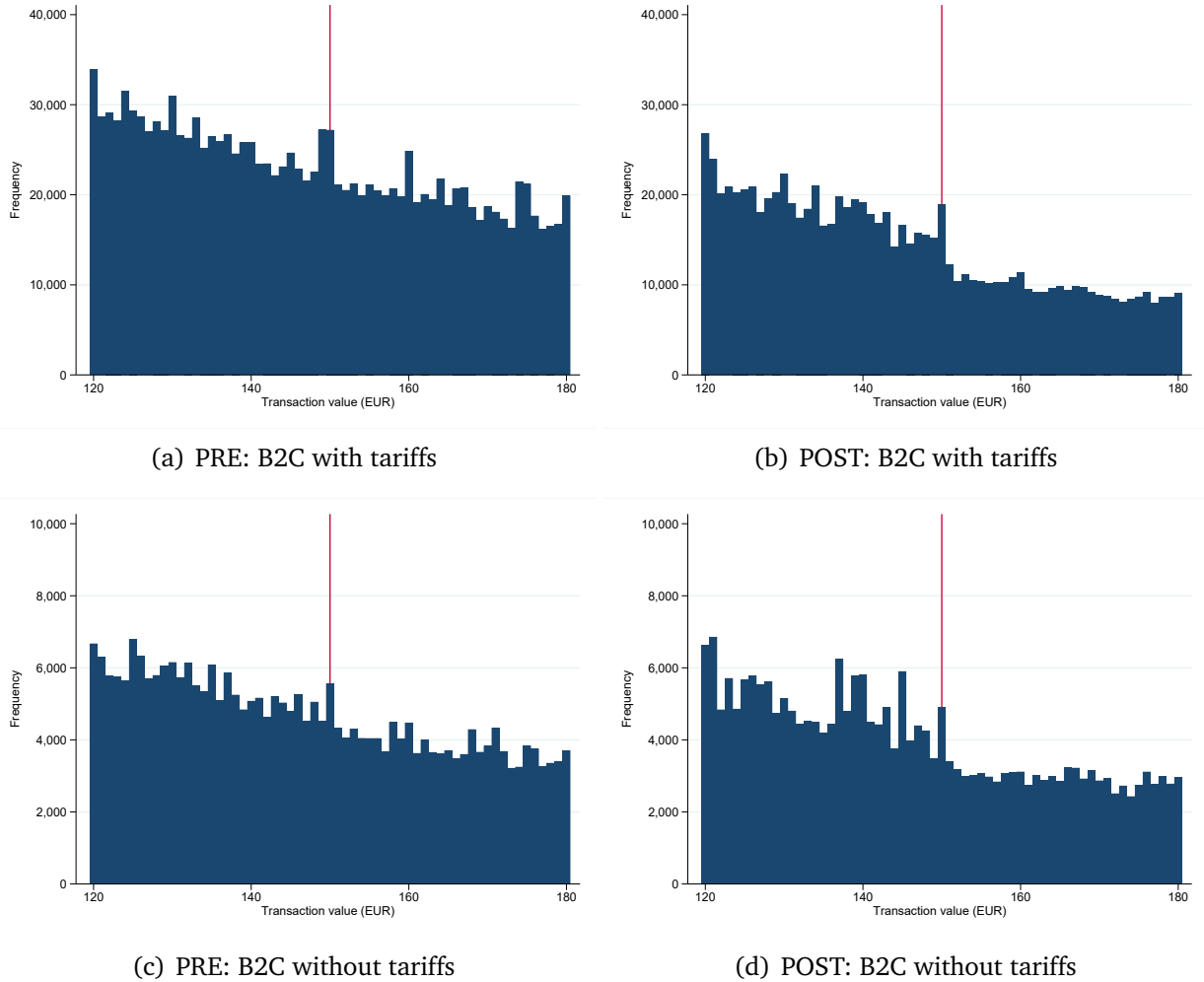
(c) PRE: B2B imports



(d) POST: B2B imports

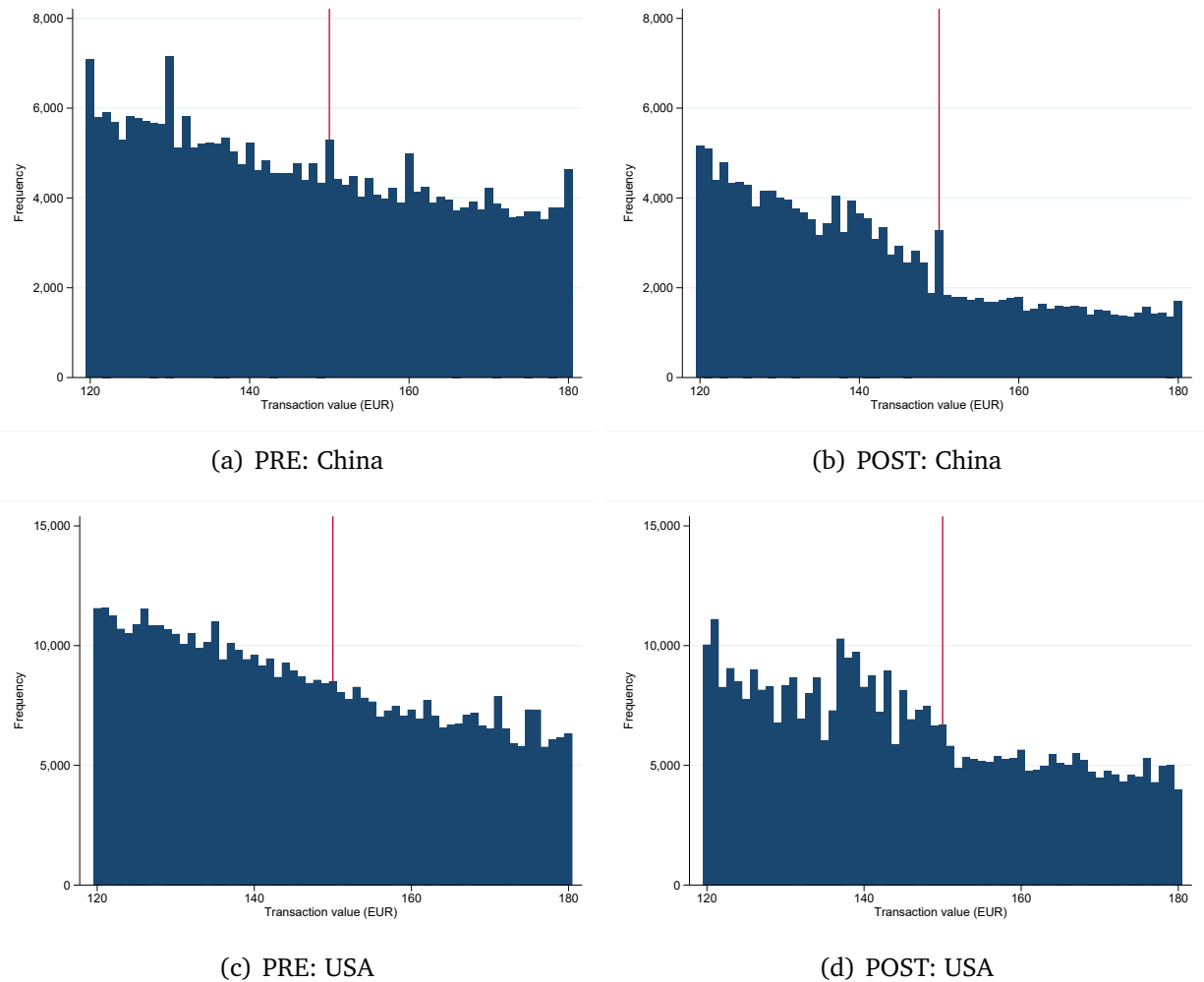
Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold, separately for B2C and B2B imports and for the pre- and post-reform. Panels (a) and (c) show the distribution of B2C and B2B imports in the pre-reform period (2018 to 2020), respectively, while Panels (b) and (d) show the corresponding distributions for the post-reform period (2022 to 2023). The sample is restricted to transactions valued between EUR 120 and EUR 180, and each bin corresponds to a EUR 1 interval. The dashed vertical line marks the EUR 150 IOSS eligibility threshold.

Figure 6: IOSS – Distribution of B2C Transaction Values by Tariff Status



Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold for B2C imports, separately by tariff status and for the pre- and post-reform periods. Panels (a) and (b) show the distribution for goods subject to tariffs in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for goods that are exempt from tariffs. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The dashed vertical line marks the EUR 150 IOSS eligibility threshold.

Figure 7: IOSS – Distribution of B2C Transaction Values from China and USA



Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold for B2C imports from China and the United States, separately for the pre- and post-reform periods. Panels (a) and (b) show the distribution of B2C imports from China in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for B2C imports from the United States. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The dashed vertical line marks the EUR 150 IOSS eligibility threshold. The corresponding B2B distributions for both countries are reported in Appendix Figure B.4.

Table 8: IOSS – Likelihood of Below-threshold Reporting

	<i>Transaction value below threshold (= 1)</i>		
	(1)	(2)	(3)
Panel A: Bandwidth EUR ± 1			
B2C x Post	0.292*** (0.069)	0.317*** (0.084)	0.113*** (0.027)
<i>Estimator</i>	Logit	Logit	PPML
<i>Product FE</i>	X		
<i>Country FE</i>		X	
<i>Product x Country FE</i>			X
<i>Year FE</i>	X	X	X
<i>Pseudo R2</i>	0.042	0.021	0.031
<i>Observations</i>	217,008	217,160	211,693
Panel B: Bandwidth EUR ± 10			
B2C x Post	0.281*** (0.037)	0.276*** (0.042)	0.108*** (0.014)
<i>Estimator</i>	Logit	Logit	PPML
<i>Product FE</i>	X		
<i>Country FE</i>		X	
<i>Product x Country FE</i>			X
<i>Year FE</i>	X	X	X
<i>Pseudo R2</i>	0.009	0.004	0.008
<i>Observations</i>	1,894,877	1,894,933	1,883,802
Panel C: Bandwidth EUR ± 30			
B2C x Post	0.399*** (0.091)	0.454*** (0.150)	0.130*** (0.031)
<i>Estimator</i>	Logit	Logit	PPML
<i>Product FE</i>	X		
<i>Country FE</i>		X	
<i>Product x Country FE</i>			X
<i>Year FE</i>	X	X	X
<i>Pseudo R2</i>	0.021	0.006	0.009
<i>Observations</i>	5,810,286	5,810,301	5,796,284

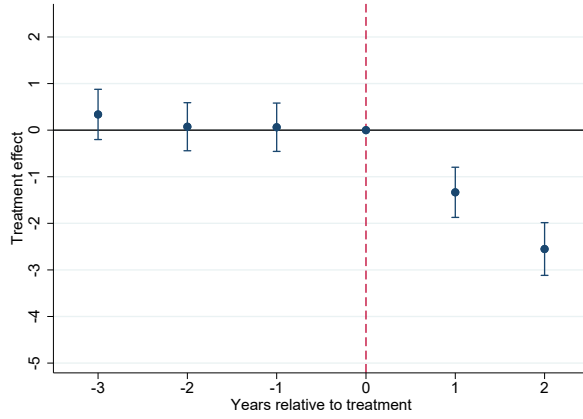
Notes: This table presents DiD estimates of the likelihood that a transaction is reported below the EUR 150 IOSS eligibility threshold, following the approach of Bernard et al. (2018). The dependent variable is an indicator equal to one if the declared transaction value falls below EUR 150 and zero otherwise, at the transaction level. The treatment group consists of B2C imports and the control group of B2B imports. Each panel restricts the sample to transactions within a different bandwidth around the threshold: EUR ± 1 in Panel A, EUR ± 10 in Panel B, and EUR ± 30 in Panel C. Columns (1) and (2) report logit estimates with product and country fixed effects, respectively, while Column (3) reports PPML estimates with product-country fixed effects. All specifications include year fixed effects. Standard errors are reported in parentheses. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

Table 9: IOSS – Effect of Simplified Reporting on Product-country Values and Weights around the Threshold

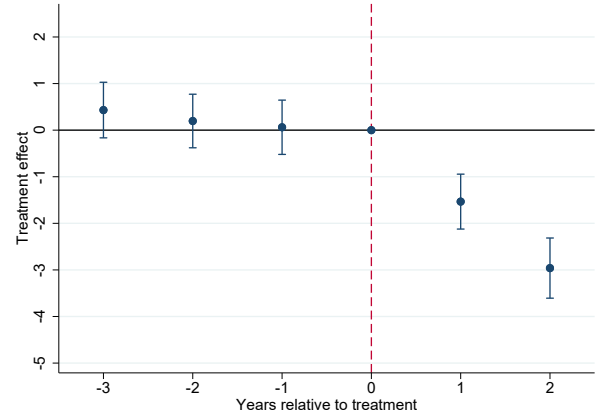
Panel A:	Product-country value (EUR)			
	Mean		Median	
	(1)	(2)	(3)	(4)
B2C x Post	-2.310*** (0.180)	-2.071*** (0.179)	-2.646*** (0.194)	-2.453*** (0.200)
<i>Product FE</i>	X		X	
<i>Product × Country FE</i>		X		X
<i>Year FE</i>	X	X	X	X
<i>Adj. R²</i>	0.02	0.10	0.02	0.09
<i>Obs</i>	104,402	92,364	104,402	92,364
Panel B:	Product-country weight (kg)			
	Mean		Median	
	(1)	(2)	(3)	(4)
B2C x Post	-0.819*** (0.178)	-0.897*** (0.161)	-0.800*** (0.184)	-0.967*** (0.169)
<i>Product FE</i>	X		X	
<i>Product × Country FE</i>		X		X
<i>Year FE</i>	X	X	X	X
<i>Adj. R²</i>	0.25	0.46	0.22	0.42
<i>Obs</i>	90,583	79,747	90,583	79,747

Notes: This table presents DiD estimates of the effect of the IOSS introduction on declared transaction values and transaction weights for B2C imports relative to B2B imports, estimated using OLS. All dependent variables are aggregated at the product-country-year level and the sample is restricted to transactions valued between EUR 120 and EUR 180. The treatment group consists of B2C imports and the control group of B2B imports. Panel A reports results for transaction values, with the dependent variable measured as either the mean or median declared transaction value in EUR per product-country-year. Panel B reports the corresponding results for transaction weights, with the dependent variable measured as either the mean or median transaction weight in kilograms per product-country-year. Observations above the 75th percentile of the weight distribution are excluded separately for B2C and B2B imports (25 kg for B2C, 95kg for B2B). Columns (1) and (3) include product and year fixed effects, while Columns (2) and (4) replace product fixed effects with product-country fixed effects. The post-reform period covers 2022 to 2023 and the pre-reform period 2018 to 2020. Standard errors are clustered at the product level. ***, **, and * denote statistical significance at the 1%, 5%, and 10% levels.

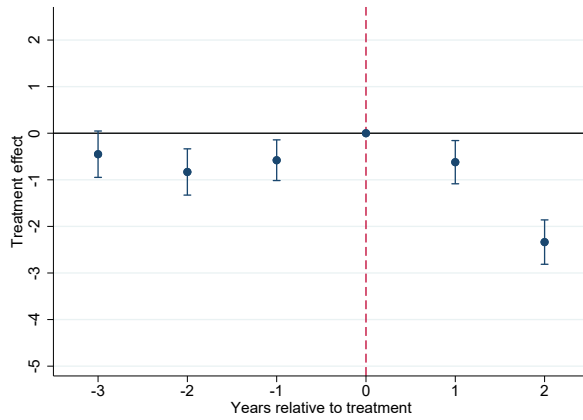
Figure 8: IOSS – Dynamic Effect of Simplified Reporting on Product-country Values and Weights around the Threshold



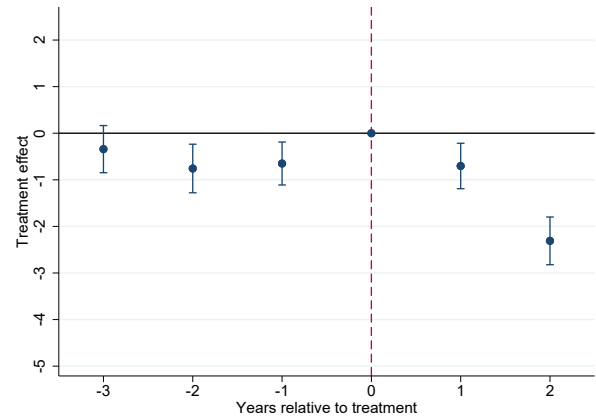
(a) Mean product-country value (EUR)



(b) Median product-country value (EUR)



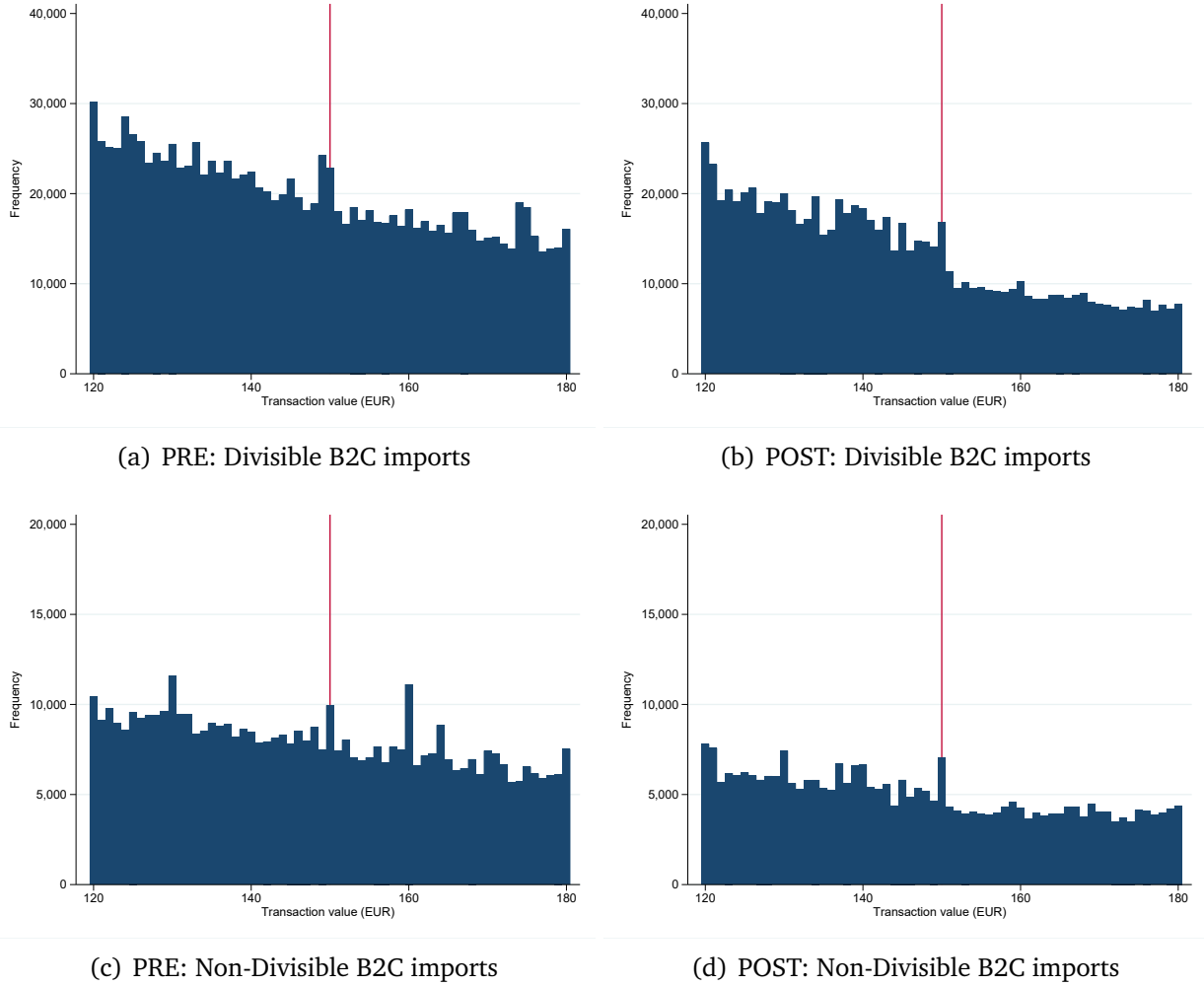
(c) Mean product-country weight (kg)



(d) Median product-country weight (kg)

Notes: This figure presents event-study estimates of the effect of the IOSS introduction on reported transaction values and transaction weights for B2C imports relative to B2B imports, obtained by replacing the *Post* indicator with separate year indicators. Panels (a) and (b) report coefficients for mean and median transaction values, respectively. Panels (c) and (d) report the corresponding coefficients for mean and median transaction weights. All dependent variables are aggregated at the product-country-year level and the sample is restricted to transactions valued between EUR 120 and EUR 180. Observations above the 75th percentile of the weight distribution are excluded separately for B2C and B2B transactions. All specifications include product-country and year fixed effects. Coefficients are plotted relative to the year of the reform (2021), marked by the dashed vertical line. Standard errors are clustered at the product level and the shaded bands denote 95% confidence intervals.

Figure 9: IOSS – Distribution of B2C Import Transaction Values for Divisible and Non-Divisible Goods



Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold for B2C imports, separately by product divisibility and for the pre- and post-reform periods. Panels (a) and (b) show the distribution for divisible goods in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for non-divisible goods. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The vertical line marks the EUR 150 IOSS eligibility threshold.

Appendix

The next pages include the following appendices:

- **OSS** (Appendix A):

Appendix A.1: Variable Definitions

Appendix A.2: Sample and Descriptives

Appendix A.3: Results

- **IOSS** (Appendix B)

Appendix B.1: Variable Definitions

Appendix B.2: Sample and Descriptives

Appendix B.3: Results

A OSS

A.1 Variable Definition OSS

Table A.1: Variable Definitions

Variable	Definition
<i>B2C</i>	Indicator variable equal to one if firm i is classified as a B2C firm, and zero otherwise.
<i>Post</i>	Indicator variable equal to one if year-month t falls after July 2021, and zero otherwise.
<i>Count</i>	Number of cross-border B2C transactions recorded by firm i in year-month t .
<i>Sales</i>	Sales volume in EUR from cross-border B2C transactions by firm i in year-month t .
<i>Goods</i>	Number of distinct product codes traded cross-border by firm i in year-month t .
<i>Countries</i>	Number of distinct EU destination countries served by firm i in year-month t .
<i>Pre mean no. employees</i>	Average number of employees of firm i per year, measured from January 2018 to July 2021 prior to the reform and held constant throughout the sample period.
<i>Pre mean total sales</i>	Average total firm sales in EUR of firm i per year, measured from January 2018 to July 2021 prior to the reform and held constant throughout the sample period.
<i>Pre mean count (DE-EU)</i>	Average number of cross-border B2C transactions recorded by firm i in year-month t , measured from January 2018 to July 2021 prior to the reform and held constant throughout the sample period.
<i>Pre mean EU-sales</i>	Average sales volume in EUR from cross-border B2C transactions by firm i in year-month t , measured from January 2018 to July 2021 prior to the reform and held constant throughout the sample period.
<i>Price</i>	Mean reported transaction value in EUR at the firm-product-year-month level.
<i>Weight</i>	Mean reported transaction weight in kilograms at the firm-product-year-month level.

A.2 Sample and Descriptives

Table A.2: OSS – Sample Selection

Selection step	Obs	Firms
Panel A: Main OSS sample		
Merged foreign transaction and VAT return data 2018–2023	594,857,060	1,687,498
Less transactions with missing industry code	(165,683,664)	(172,959)
Less import transactions	(158,890,992)	(1,018,092)
Less extra-EU transactions	(98,266,882)	(42,548)
Less transactions to Great Britain (Brexit)	(1,519,474)	(95)
Less transactions subject to excise duties	(1,388,292)	(59)
Less transactions subject to MOSS	(5,518,776)	(28,397)
Less singleton firms	(77,344)	(77,344)
OSS sample (firm year-month panel)	8,628,679	348,004
Panel B: Treatment intensity sample		
Merged foreign transaction and VAT return data 2018–2023	594,857,060	1,687,498
Less transactions with missing industry code	(165,683,664)	(172,959)
Less import transactions	(158,890,992)	(1,018,092)
Less transactions to Great Britain (Brexit)	(9,049,532)	(1,070)
Less transactions subject to excise duties	(2,232,504)	(597)
Less transactions subject to MOSS	(7,726,309)	(30,852)
Less firms classified as B2B	(248,802,007)	(308,056)
Less non-missing control variables	(915,486)	(91,030)
Less singleton firms	(9,461)	(9,461)
OSS treatment intensity sample (firm year-month panel)	1,547,105	55,381

Notes: This table reports the construction of the samples used for the OSS analysis. Panel A reports the sample selection for the main sample including B2C and B2B firms selling within the EU. Panel B reports the sample selection for the treatment intensity sample which contains only B2C firms selling to consumers in the EU as well as in third countries.

Table A.3: OSS – B2C Classification

Code	Industry	Code	Industry
45202	Car wash facility.	71111	Architectural offices for building construction.
45320	Retail sale of motor vehicle parts and accessories.	47190	Other retail sale of various types of goods.
47110	Retail sale of food, luxury foods, and beverages.	47210	Retail sale of fruit, vegetables, and potatoes.
47220	Retail sale of meat and meat products.	47230	Retail sale of fish, seafood, and fish products.
47240	Retail sale of baked goods and confectionery.	47250	Retail sale of beverages.
47260	Retail sale of tobacco products.	47290	Other retail sale of food and beverages.
47301	Retail sale of motor fuels on behalf of third parties (agency gas stations).	47302	Retail sale of motor fuels under own name (independent gas stations).
47410	Retail sale of data processing equipment, peripheral devices, software.	47420	Retail sale of telecommunications equipment.
47430	Retail sale of consumer electronics equipment.	47510	Retail sale of textiles.
47521	Retail sale of metal and plastic goods n.e.c..	47523	Retail sale of paints, building materials, and DIY supplies.
47530	Retail sale of curtains, carpets, floor coverings, and wallpaper.	47540	Retail sale of electrical household appliances.
47591	Retail sale of home furniture.	47592	Retail sale of ceramic products and glassware.
47593	Retail sale of musical instruments and music supplies.	47599	Retail sale of household goods n.e.c..
47610	Retail sale of books.	47621	Retail sale of magazines and newspapers.
47622	Retail sale of stationery, paper goods, school and office supplies.	47630	Retail sale of recorded audio and video media.
47641	Retail sale of bicycles, bicycle parts, and accessories.	47642	Retail sale of sports and camping goods (excluding camping furniture).
47650	Retail sale of toys.	47710	Retail sale of clothing.
47721	Retail sale of shoes.	47722	Retail sale of leather goods and luggage.
47730	Pharmacies.	47740	Retail sale of medical and orthopedic goods.
47750	Retail sale of cosmetic products and personal care products.	47761	Retail sale of flowers, plants, seeds, and fertilizers.
47762	Retail sale of zoological supplies and live animals.	47770	Retail sale of watches and jewelry.
47781	Opticians.	47782	Retail sale of photographic and optical goods (excluding opticians).
47783	Retail sale of art, pictures, decorative arts, stamps, coins, and gift items.	47789	Other retail sale n.e.c. (in sales rooms).
47791	Retail sale of antiques and antique carpets.	47792	Antique shops.
47799	Retail sale of other second-hand goods.	47810	Retail sale of food, beverages, tobacco products at stalls and markets.
47820	Retail sale of textiles, clothing, and footwear at stalls and markets.	47890	Retail sale of other goods at stalls and markets.
47910	Mail order and internet retail.	47991	Retail sale of fuel from warehouses.
47999	Other retail sale n.e.c. (not in sales rooms).	49100	Long-distance passenger transport by rail.
49310	Local passenger transport by land (excluding taxis).	49320	Operation of taxis.
49391	Long-distance passenger transport by bus.	49392	Occasional passenger transport by bus.
49399	Passenger transport by land, n.e.c..	49410	Freight transport by road.
49420	Removal services.	50100	Sea and coastal passenger water transport.
50300	Passenger transport by inland waterway.	51100	Passenger transport by air.
52211	Operation of parking garages and parking lots.	52214	Operation of stations for passenger transport, including bus stations.
55101	Hotels (excluding bed and breakfast hotels).	55102	Bed and breakfast hotels.
55103	Inns.	55104	Guesthouses.
55201	Recreation and vacation homes.	55202	Vacation centres.
55203	Vacation homes and vacation apartments.	55204	Youth hostels and cabins.
55300	Campgrounds.	55901	Private accommodations.
55909	Other accommodation facilities n.e.c..	56101	Restaurants mit herkömmlicher Bedienung.
56102	Self-service restaurants.	56103	Snack bars and similar establishments.
56104	Cafés.	56105	Ice cream parlors.
56106	Pizzerias.	56107	Restaurants serving Asian food.
56108	Snack bars serving Asian food.	56109	Kebab snack bars.
56210	Event caterers.	56290	Provision of other catering services.
56301	Pubs.	56302	Discotheques and dance halls.
56303	Bars.	56304	Amusement venues.
56309	Other beverage-oriented catering establishments.	58110	Publishing of books.
58120	Publishing of directories and mailing lists.	58130	Publishing of newspapers.
58140	Publishing of magazines.	58190	Other publishing activities (excluding software).
58210	Publishing of computer games.	58290	Publishing of other software.
59110	Production of films, video films, and television programs.	59111	Social media players' advertising revenue on social networks.
59119	Production of films, video films, and television programs n.e.c..	59120	Post-production and other film technology.
59130	Film distribution and sales (excluding video stores).	59140	Cinemas.
59201	Sound studios and production of radio programs.	59202	Publishing of recorded sound carriers.
59203	Publishing of sheet music.	60100	Radio broadcasters.
60200	Television broadcasters.	61100	Wired telecommunications.
61200	Wireless telecommunications.	61300	Satellite telecommunications.
61901	Internet service providers.	61909	Other telecommunications n.e.c..
64110	Central banking.	64191	Credit banks, including branches of foreign banks.
64192	Credit institutions in the savings bank sector.	64193	Credit institutions in the cooperative sector.
64194	Mortgage banks.	64195	Credit institutions with special tasks.
64196	Building societies.	64201	Venture capital funds.
64202	Private equity funds.	64203	Mezzanine funds.
64204	Investment companies.	64300	Trust and other funds and similar financial institutions.
64910	Financial leasing institutions.	64921	Special credit institutions (excluding pawnbroking).
64922	Pawnshops.	64991	Equity funds.
64992	Mixed funds.	64993	Real estate funds.
64994	Other investment funds.	64995	Special real estate funds.
64996	Other special funds.	64999	Other financial institutions n.e.c..
65110	Life insurance companies.	65121	Health insurance.
65122	Property and casualty insurance.	65200	Reinsurance.
65300	Pension funds and pension schemes.	66110	Securities and commodities exchanges.

Code	Industry	Code	Industry
66120	Securities and commodities trading.	66190	Other activities related to financial services.
66191	Payment institutions.	66199	Other activities related to financial services.
66210	Risk and damage assessment.	66220	Activities of insurance agents and brokers.
66290	Other activities related to insurance services and pension funds.	66300	Fund management activities.
68100	Purchase and sale of own land, buildings, and apartments.	68200	Rental and leasing of own or leased land, buildings, and apartments.
68203	REIT (Real Estate Investment Trust).	68310	Brokerage of land, buildings, and apartments for third parties.
68320	Management of land, buildings, and apartments for third parties.	69101	Law firms with notary offices.
69102	Law firms without notary offices.	69103	Notary offices.
69104	Patent attorney offices.	69109	Provision of other legal services n.e.c..
69203	Tax agents, tax consultants, and tax consulting firms.	69204	Accounting (excluding data processing services).

Notes: This table lists the five-digit industry codes and corresponding industry descriptions used to classify firms as B2C in our baseline specification. Firms in all remaining industry codes are assigned to the control group and classified as B2B.

Table A.4: OSS – MOSS classification

Code	Industry
26200	Manufacture of data processing equipment and peripheral devices.
26300	Manufacture of telecommunications equipment and devices.
46510	Wholesale of data processing equipment, peripheral devices, and software.
46520	Wholesale of electronic components and telecommunications equipment..
47410	Retail sale of data processing equipment, peripheral devices, and software.
47420	Retail sale of telecommunications equipment.
58210	Publishing of computer games.
58290	Publishing of other software.
59110	Production of films, video films, and television programs.
59111	Social media players that generate advertising revenue on social networks.
59119	Production of films, video films, and television programs n.e.c..
59201	Sound studios and production of radio programs.
60100	Radio broadcasters.
60200	Television broadcasters.
61100	Wired telecommunications.
61200	Wireless telecommunications.
61300	Satellite telecommunications.
61901	Internet service providers.
61909	Other telecommunications n.e.c..
62011	Development and programming of Internet presentations.
62019	Other software development.
62020	Provision of consulting services in the field of information technology.
62030	Operation of data processing facilities for third parties.
62090	Provision of other information technology services.
63110	Data processing, hosting and related activities.
63120	Web portals.
63910	Correspondence and news agencies.
63990	Provision of other information services n.e.c..
82200	Call centres.
90014	Self-employed stage, film, radio, and television artists, as well as other performing artists.
95110	Repair of data processing equipment and peripheral devices.
95120	Repair of telecommunications equipment.

Notes: This table lists the five-digit industry codes and corresponding industry descriptions used to identify firms eligible for the Mini-One-Stop-Shop (MOSS) regime.

Table A.5: OSS – Top B2C Intra-EU Destination Countries

B2C firms					
Panel A: Country	<i>Firms number</i>	<i>Count number</i>	<i>Volume value (EUR)</i>	<i>Mean value (EUR)</i>	<i>Median value (EUR)</i>
Austria	6,016	2,127,813	11,091,441,664	5,212.60	218.00
Netherlands	5,326	1,186,884	7,805,392,896	6,576.37	230.00
France	4,495	1,129,626	7,656,053,760	6,777.51	212.00
Poland	4,358	829,006	16,980,992,000	20,483.56	328.00
Italy	4,326	780,689	5,680,328,192	7,276.04	196.00
Czechia	3,893	754,261	4,286,220,800	5,682.68	225.00
Belgium	3,967	693,166	2,965,968,384	4,278.87	173.00
Spain	3,303	529,268	3,030,808,832	5,726.42	178.00
B2C firms					
Panel B: Type of good	<i>Firms number</i>	<i>Count number</i>	<i>Volume value (EUR)</i>	<i>Mean value (EUR)</i>	<i>Median value (EUR)</i>
99 Other commodities	106,567	1,299,660	12,426,172,416	9,561.09	1,580.00
85 Electrical machinery	1,798	1,294,393	8,522,830,848	6,584.42	193.00
84 Machinery & Mech. Appliances	1,722	872,629	7,952,932,864	9,113.76	306.00
61 Clothing (knitted or croch.)	832	861,466	5,772,258,304	6,700.51	237.00
62 Clothing (not knitted or croch.)	762	802,918	6,365,382,144	7,927.81	301.00
39 Plastics	1,853	691,069	1,835,319,296	2,655.77	105.00
82 Base Metal Products	825	479,761	700,117,824	1,459.31	97.00
73 Iron or Steel	1,379	475,054	994,636,736	2,093.73	97.00

Notes: This table reports the eight most frequent destination countries and product categories for cross-border B2C sales by German firms in the pre-reform period from January 2018 to June 2021, ranked by the total number of transactions in each panel. Panel A reports the eight most frequent destination countries, ranked in descending order by transaction count. Panel B reports the eight most frequently exported product categories, identified using the eight-digit Combined Nomenclature (CN) product classification and aggregated to the two-digit level for presentation. In both panels, the data are sourced from the German foreign trade statistics and are restricted to intra-EU B2C exports by firms classified as B2C based on the five-digit industry classification. Full product group descriptions are provided in Appendix Table A.6.

Table A.6: OSS: Frequently Exported Goods to Final Customers in the EU
with Full Product Group Name

Code	Product group
39	Plastics and articles thereof.
61	Articles of apparel and clothing accessories, knitted or crocheted.
62	Articles of apparel and clothing accessories, not knitted or crocheted.
73	Articles of iron or steel.
82	Tools, implements, cutlery, spoons and forks, of base metal; parts thereof of base metal.
84	Nuclear reactors, boilers, machinery and mechanical appliances; parts thereof.
85	Electrical machinery and equipment and parts thereof; sound recorders and reproducers, television image and sound recorders and reproducers, and parts and accessories of such articles.
94	Furniture; bedding, mattresses, mattress supports, cushions and similar stuffed furnishings; lamps and lighting fittings, not elsewhere specified or included; illuminated signs, illuminated nameplates and the like; prefabricated buildings.

Notes: This table provides the full product group descriptions for the exported goods from German B2C firms to EU final customers reported in the main analysis. Product descriptions are based on the TARIC nomenclature and were accessed via the EU TARIC database at https://ec.europa.eu/taxation_customs/dds2/taric/taric_consultation.jsp?Lang=en (accessed April 2026).

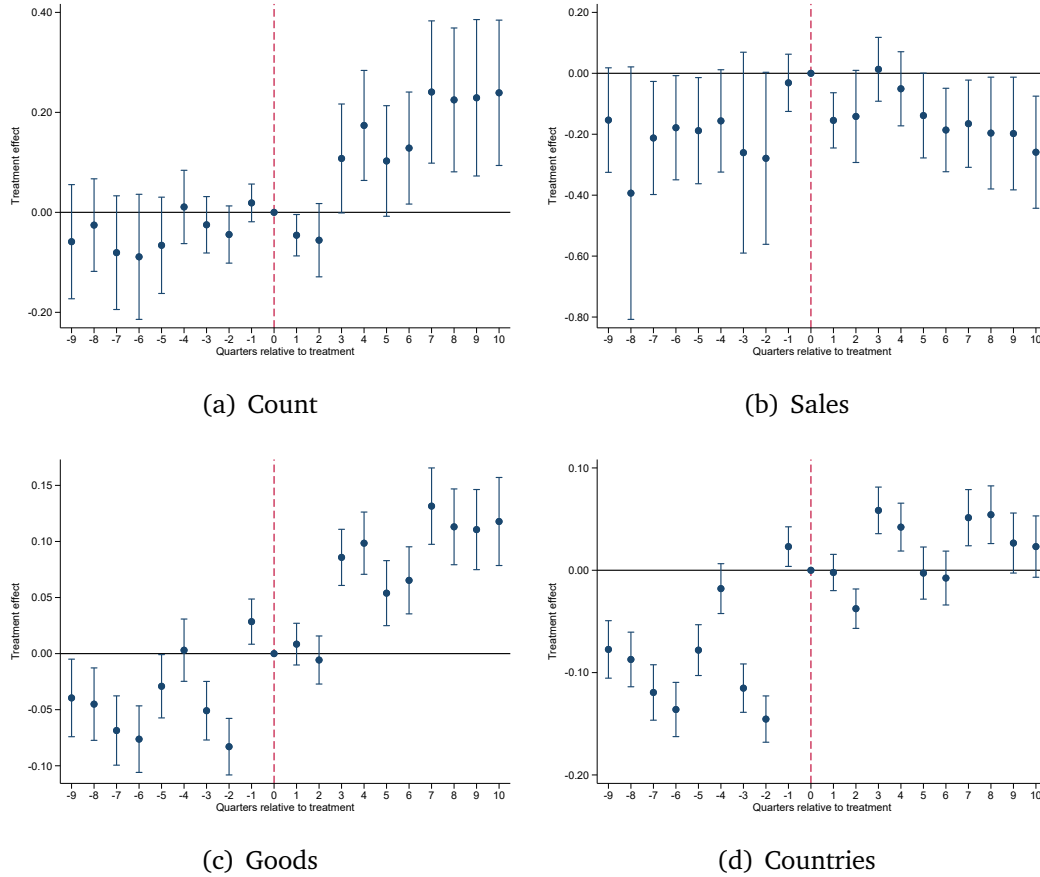
Table A.7: OSS – Descriptive Statistics (Treatment Intensity)

Panel A:		Intra EU B2C firms				
	<i>Mean</i>	<i>SD</i>	<i>P1</i>	<i>P50</i>	<i>P99</i>	<i>Obs</i>
<i>Dependent variables:</i>						
Count	19.60	409.65	0.00	1.00	269	1,139,954
Sales (EUR)	103,800.48	2,742,257.79	0.00	2,077.00	1,061,509	1,139,954
Goods	1.64	3.92	0.00	1.00	20	1,139,954
Countries	1.75	3.36	0.00	1.00	20	1,139,954
<i>Control variables:</i>						
Pre mean no. employees	41.39	655.14	0.00	1.32	551	1,135,639
Pre mean total sales (mio. EUR)	13.66	238.85	0.03	0.66	127.49	1,135,639
Pre mean count (DE-EU)	16.93	297.29	1.00	1.00	231	1,139,954
Pre mean EU-sales (mio. EUR)	0.09	2.36	0.00	0.00	0.90	1,139,954
<i>No. of firms</i>						43,593
Panel B:		Third country B2C firms				
	<i>Mean</i>	<i>SD</i>	<i>P1</i>	<i>P50</i>	<i>P99</i>	<i>Obs</i>
<i>Dependent variables:</i>						
Count	8.25	80.54	0.00	1.00	108	407,151
Sales (EUR)	70,572.86	765,295.02	0.00	5,470.00	1,123,891	407,151
Goods	2.06	3.96	0.00	1.00	18	407,151
Countries	1.31	2.20	0.00	1.00	9	407,151
<i>Control variables:</i>						
Pre mean no. employees	61.34	696.85	0.00	3.31	771	402,051
Pre mean total sales (mio. EUR)	10.07	160.69	0.04	1.17	106.38	402,051
Pre mean count (DE-EU)	8.58	61.18	1.00	2.67	96	407,151
Pre mean EU-sales (mio. EUR)	0.07	0.41	0.00	0.01	0.94	407,151
<i>No. of firms</i>						11,788

Notes: This table reports descriptive statistics for the alternative treatment intensity sample, covering the period January 2018 to December 2023. The data combine the German foreign trade statistics with VAT return data and are aggregated at the firm-month level, such that each observation represents a firm's total cross-border activity in a given month. The sample is restricted to B2C firms, with treated firms primarily exporting to EU member states and control firms primarily exporting to non-EU countries. Panel A reports statistics for treated firms and Panel B for control B2C firms. Variable definitions are provided in Appendix Table A.1.

A.3 Results

Figure A.1: OSS – Dynamic Effect of Simplified Reporting on Business Activity (Treatment Intensity)



Notes: This figure presents event-study estimates of the effect of the OSS introduction on cross-border business activity using the alternative treatment intensity sample, obtained by replacing the *Post* indicator in Equation 1 with separate year-month indicators. To display the full post-reform period, the monthly firm-level panel is aggregated to the firm quarter level. The treatment group consists of firms primarily exporting to EU member states and the control group of firms primarily exporting to non-EU countries. The specification includes firm and year quarter fixed effects and pre-reform cross-border trade controls interacted with a linear time trend. Coefficients are plotted relative to the quarter immediately preceding the reform (Q2 2021), marked by the dashed vertical line. Standard errors are clustered at the firm level and the blue bands denote 95% confidence intervals.

B IOSS

B.1 Variable Definition IOSS

Table B.1: Variable Definitions

Variable	Definition
<i>B2C</i>	Indicator variable equal to one if transaction s is classified as a B2C import, and zero otherwise.
<i>Post</i>	Indicator variable equal to one if transaction s occurs in the post-reform period (2022 to 2023), and zero otherwise.
<i>Count</i>	Number of import transactions aggregated at the product-country-year level.
<i>Sales</i>	Total reported value in EUR of import transactions aggregated at the product-country-year level.
<i>Below threshold</i>	Indicator variable equal to one if the reported transaction value of import s is below the EUR 150 IOSS eligibility threshold, and zero otherwise.
<i>Transaction value</i>	Reported net value of import transaction s in EUR.
<i>Transaction weight</i>	Reported weight of import transaction s in kilograms.

B.2 Sample and Descriptives

Table B.2: IOSS – Sample Selection

Selection step	Obs
Raw customs transaction data 2018–2023 (ATLAS and ATLAS Impost)	264,320,357
Less transactions with EU origin or EU dispatch country	17,211,794
Less non-commercial and special transactions (private, gifts, special arrangements)	20,429,998
Less transactions originating from Great Britain (Brexit)	62,389,228
Less transactions below the de minimis threshold of EUR 30	93,257,865
Less transactions recorded during the transitional year 2021	11,807,645
Less transactions subject to excise duties	1,057,153
Less transactions with missing or invalid country or product codes	1,658
IOSS transaction year sample (EUR 30 - 1,500)	58,165,673
IOSS transaction year sample (EUR 120 - 180)	5,803,263
IOSS product-country year sample (EUR 30 - 150)	178,135
IOSS product-country year sample (EUR 120 - 180)	104,402

Notes: This table reports the construction of the IOSS sample.

Table B.3: IOSS – Top B2C Import Countries

Panel A: B2C Transactions between EUR 30 and EUR 150					
Country	<i>Count number (Tsd.)</i>	<i>Share %</i>	<i>Volume value (Mio. EUR)</i>	<i>Mean value (EUR)</i>	<i>Median value (EUR)</i>
United States	8,190.25	30.48	529.20	64.61	54
China	6,090.21	22.66	353.51	58.05	49
Switzerland	2,868.03	10.67	191.47	66.76	56
Turkey	2,671.43	9.94	150.15	56.21	46
Vietnam	1,394.34	5.19	78.21	56.09	54
Hong Kong	927.12	3.45	74.49	80.35	79
Japan	845.39	3.15	60.64	71.73	63
India	463.20	1.72	31.53	68.08	59
Panel B: B2C Transactions above EUR 150					
Country	<i>Count number (Tsd.)</i>	<i>Share %</i>	<i>Volume value (Mio. EUR)</i>	<i>Mean value (EUR)</i>	<i>Median value (EUR)</i>
United States	2,814.84	26.15	1,177.86	423.41	289
China	2,186.30	20.31	1,215.06	558.12	438
Switzerland	1,706.67	15.85	849.02	498.10	362
Hong Kong	685.35	6.37	330.00	481.91	343
Turkey	599.75	5.57	319.74	534.97	401
Japan	519.87	4.83	229.24	461.56	327
India	279.43	2.60	142.06	548.44	402
Taiwan	200.57	1.86	119.59	606.21	498

Notes: This table reports the eight most frequent non-EU source countries for imports to final customers in Germany in the pre-reform period 2018 to 2020, ranked by the total number of transactions in each panel. Panel A covers B2C transactions valued between EUR 30 and EUR 150, Panel B covers B2C transactions valued between EUR 150 and EUR 1,500.

Table B.4: IOSS – Top B2C Import Products

Panel A: Transactions between EUR 30 and EUR 150						
Type of good		<i>Count</i> <i>number (Tsd.)</i>	<i>Share</i> <i>%</i>	<i>Volume</i> <i>value (Mio. EUR)</i>	<i>Mean</i> <i>value (EUR)</i>	<i>Median</i> <i>value (EUR)</i>
61	Clothing (knitted)	6,598.22	24.70	387.00	58.65	49
62	Clothing (not knitted)	4,520.63	16.92	260.45	57.61	48
64	Footwear	2,953.15	11.05	180.16	61.01	54
85	Electrical machinery	1,219.04	4.56	86.41	70.88	62
39	Plastics	1,156.76	4.33	81.00	70.02	61
95	Toys, games, sports	1,142.60	4.28	79.01	69.15	60
49	Books and newspapers	942.31	3.53	62.16	65.96	56
71	Jewellery	867.62	3.25	58.51	67.44	59
Panel B: Transactions above EUR 150						
Type of good		<i>Count</i> <i>number (Tsd.)</i>	<i>Share</i> <i>%</i>	<i>Volume</i> <i>value (Mio. EUR)</i>	<i>Mean</i> <i>value (EUR)</i>	<i>Median</i> <i>value (EUR)</i>
61	Clothing (knitted)	1,125.68	10.46	462.91	413.97	275
39	Plastics	999.18	9.28	545.48	548.44	428
85	Electrical machinery	939.81	8.73	473.67	505.14	380
62	Clothing (not knitted)	795.63	7.39	341.52	431.65	295
87	Vehicles	616.24	5.72	326.46	530.43	395
95	Toys, games, sports	529.34	4.92	222.96	423.86	292
42	Leather goods	436.95	4.06	195.35	448.90	304
64	Footwear	413.22	3.84	137.88	334.99	235

Notes: This table reports the eight most frequently imported product categories from non-EU countries to final customers in Germany in the pre-reform period 2018 to 2020, ranked by the total number of transactions in each panel. Panel A covers B2C transactions valued between EUR 30 and EUR 150, Panel B covers B2C transactions valued between EUR 150 and EUR 1,500. Full product group descriptions are provided in Appendix Table B.5.

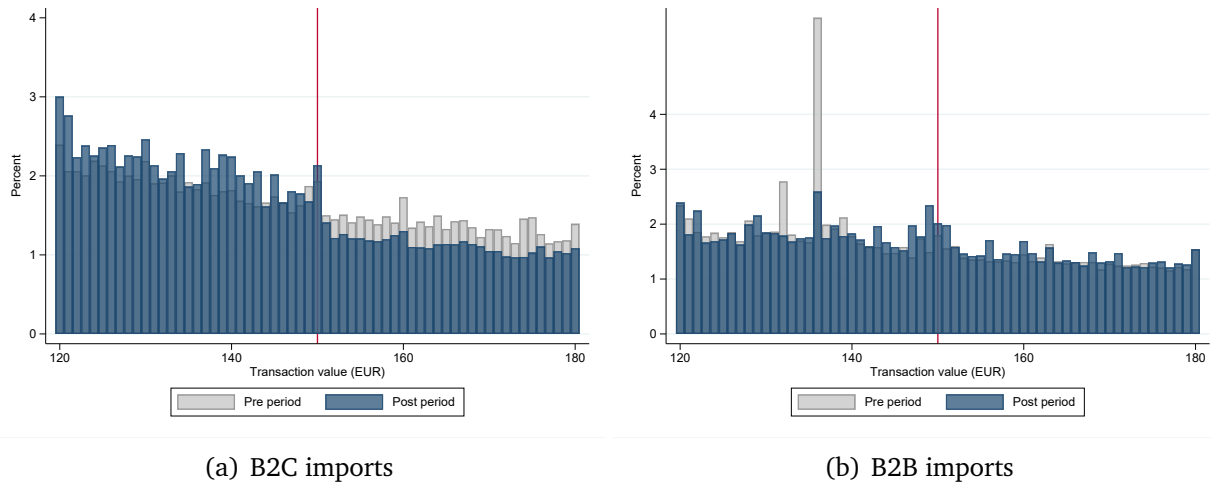
Table B.5: IOSS – Frequently imported goods with full product group name

Code	Product group name
39	Plastics and articles thereof.
42	Articles of leather; saddlery and harness; travel goods, handbags and similar containers, articles of animal gut.
49	Printed books, newspapers, pictures and other products of the printing industry; manuscripts, typescripts and plans.
61	Articles of apparel and clothing accessories, knitted or crocheted.
62	Articles of apparel and clothing accessories, not knitted or crocheted.
64	Footwear, gaiters and the like; parts of such articles.
71	Natural or cultured pearls, precious or semi-precious stones, precious metals, metals clad with precious metal, and articles thereof; imitation jewellery; coin.
85	Electrical machinery and equipment and parts thereof; sound recorders and reproducers, television image and sound recorders and reproducers, and parts and accessories of such articles.
87	Vehicles other than railway or tramway rolling stock, and parts and accessories thereof.
90	Optical, photographic, cinematographic, measuring, checking, precision, medical or surgical instruments and apparatus; parts and accessories thereof.
95	Toys, games and sports requisites; parts and accessories thereof.

Notes: This table provides the full product group descriptions for the most frequently imported goods from non-EU countries to final customers in Germany reported in Appendix Table B.4. Product descriptions are based on the TARIC nomenclature and were accessed via the EU TARIC database at https://ec.europa.eu/taxation_customs/dds2/taric/taric_consultation.jsp?Lang=en (accessed April 2026).

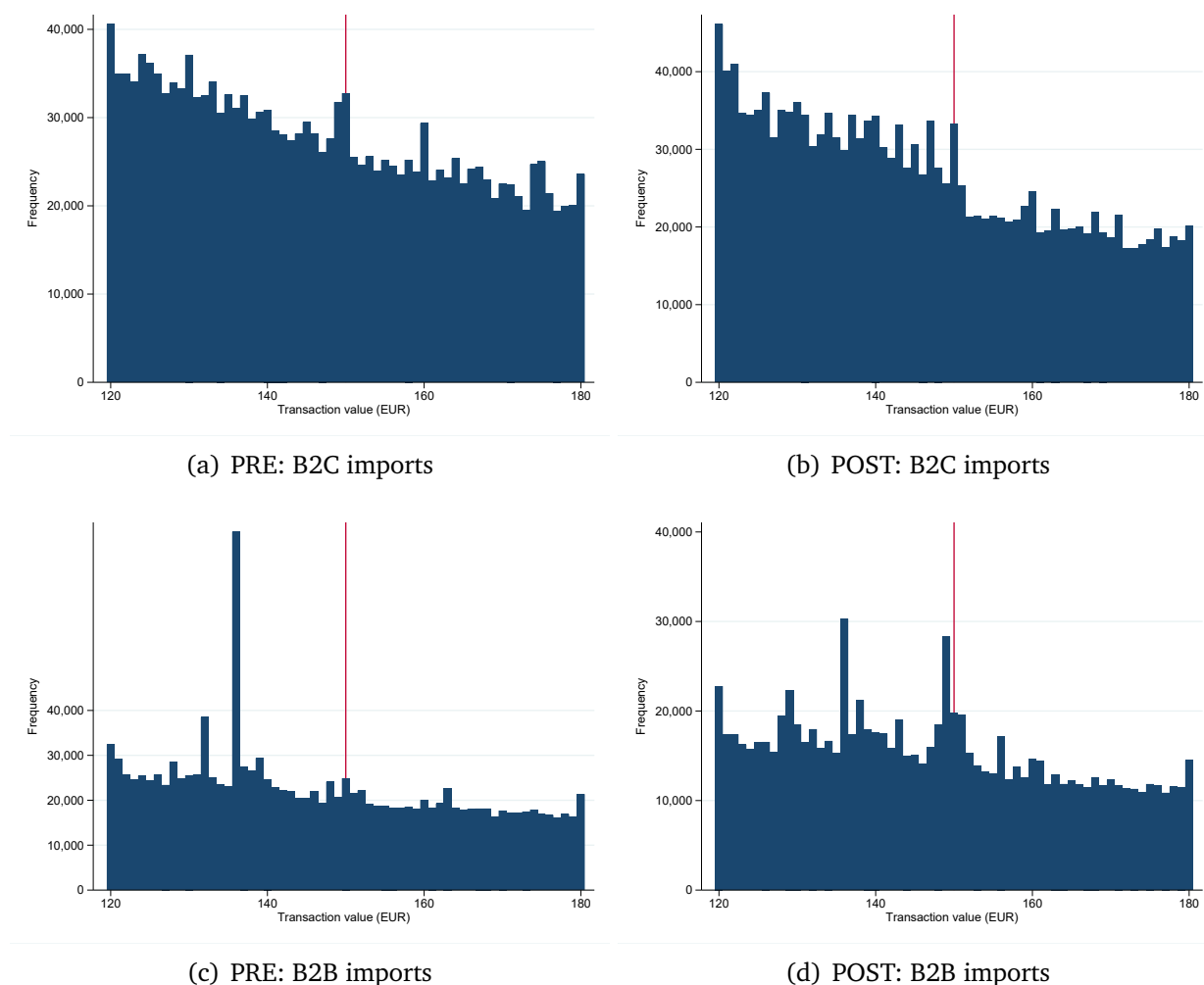
B.3 Results

Figure B.1: IOSS – Distribution of B2C and B2B Transaction Values in Percent



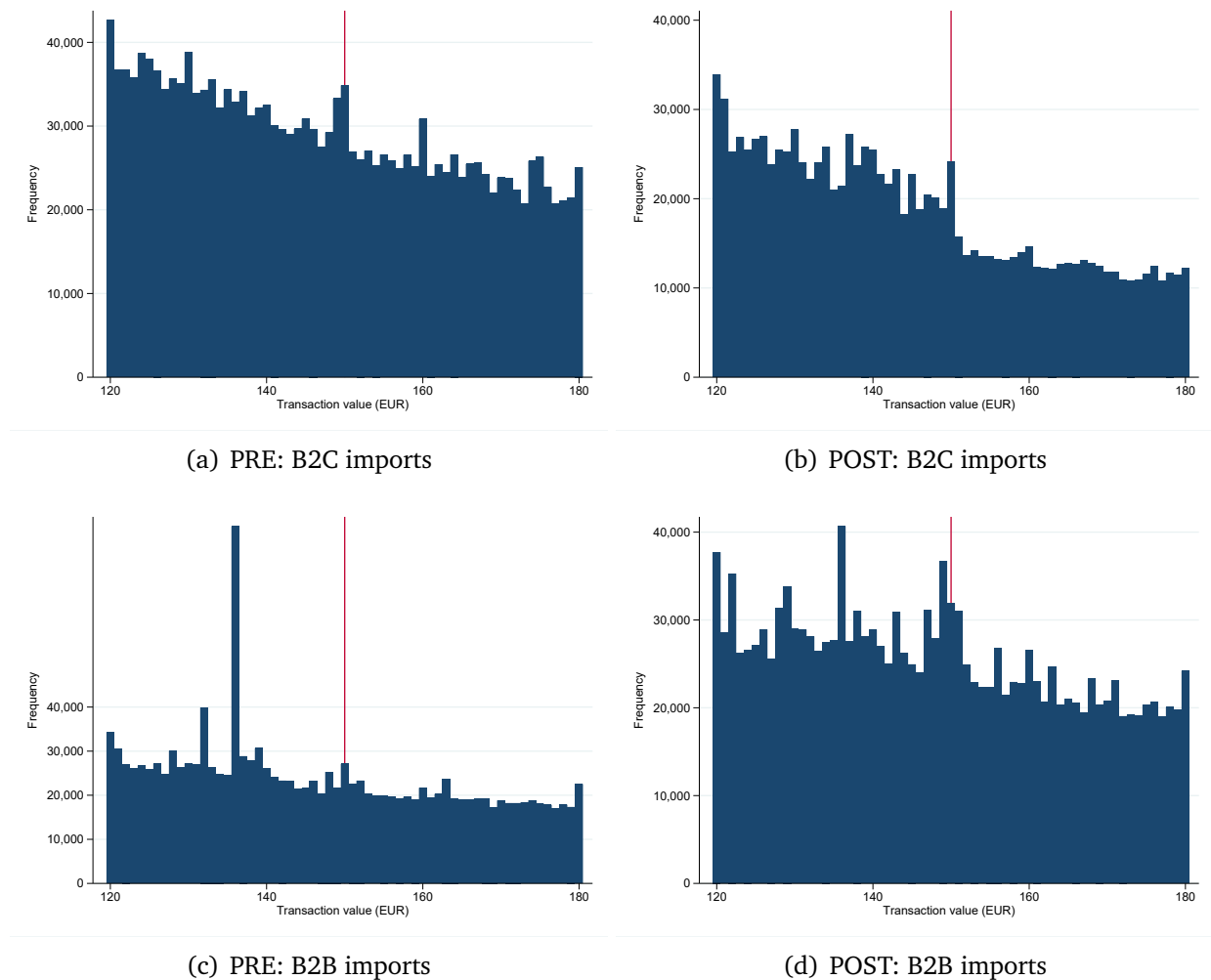
Notes: This figure shows histograms of the percent of transaction values for B2C and B2B imports before and after the introduction of the IOSS. Panel (a) displays the distribution of B2C imports in the pre-reform period (2018–2020) and post-reform period (2022–2023), respectively. Panel (b) shows the corresponding distributions for B2B imports. The distribution is restricted to transactions valued between EUR 120 and EUR 180, with the IOSS eligibility threshold marked at EUR 150.

Figure B.2: IOSS – Distribution of B2C and B2B Transaction Values using a Strict Product-based B2C Classification



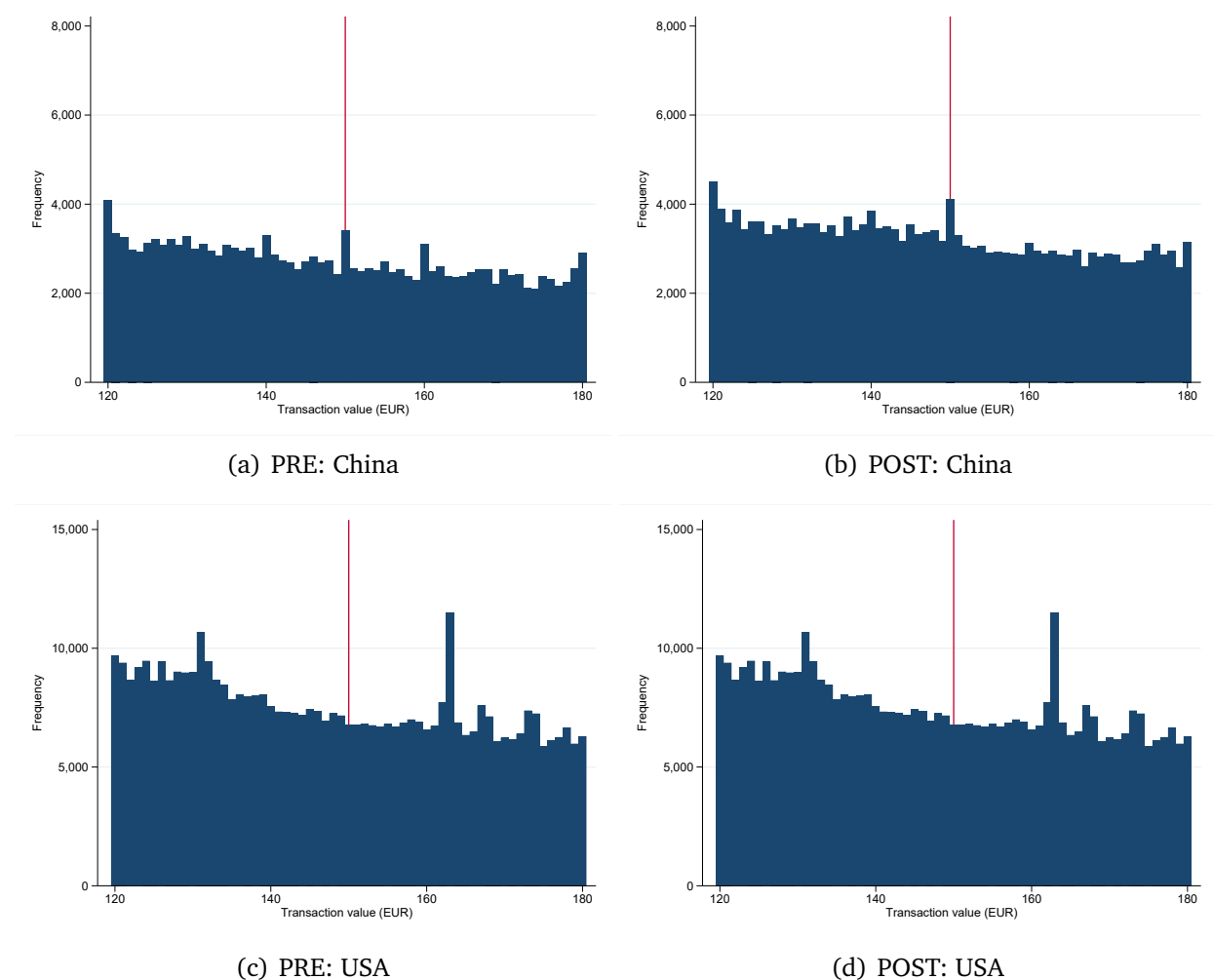
Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold, separately for B2C and B2B imports and for the pre- and post-reform using a strict-product based B2C classification, in which each transaction is assigned to the same B2C or B2B category in both the pre- and post-reform periods. Panels (a) and (b) display the distribution of B2C imports in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for B2B imports. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The vertical line marks the EUR 150 IOSS eligibility threshold.

Figure B.3: IOSS – Distribution of B2C and B2B Transaction Values using a Looser B2C Classification



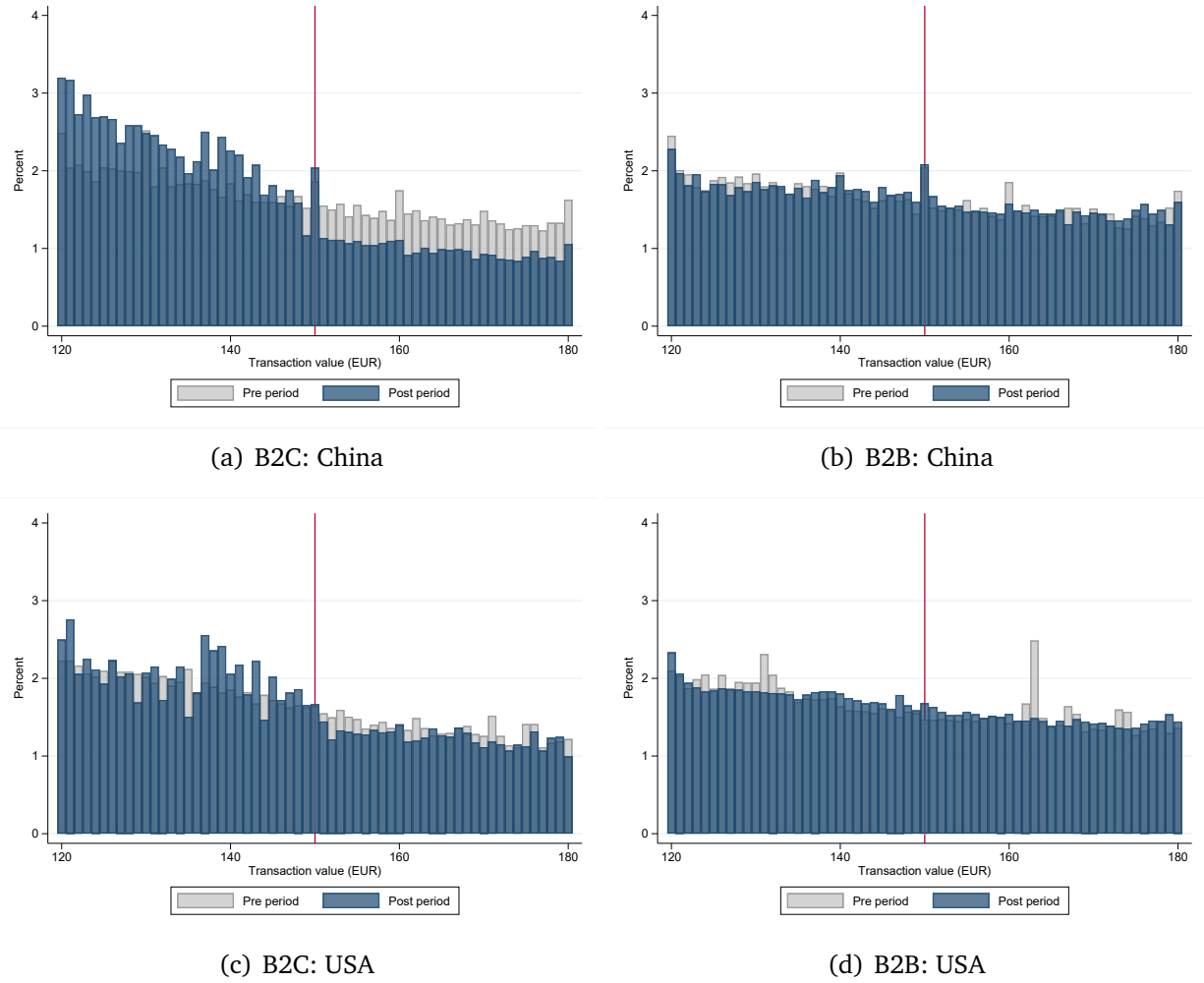
Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold, separately for B2C and B2B imports and for the pre- and post-reform using a looser B2C classification criterion. Panels (a) and (b) display the distribution of B2C imports in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for B2B imports. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The vertical line marks the EUR 150 IOSS eligibility threshold.

Figure B.4: IOSS – Distribution of B2B Transaction Values from China and USA



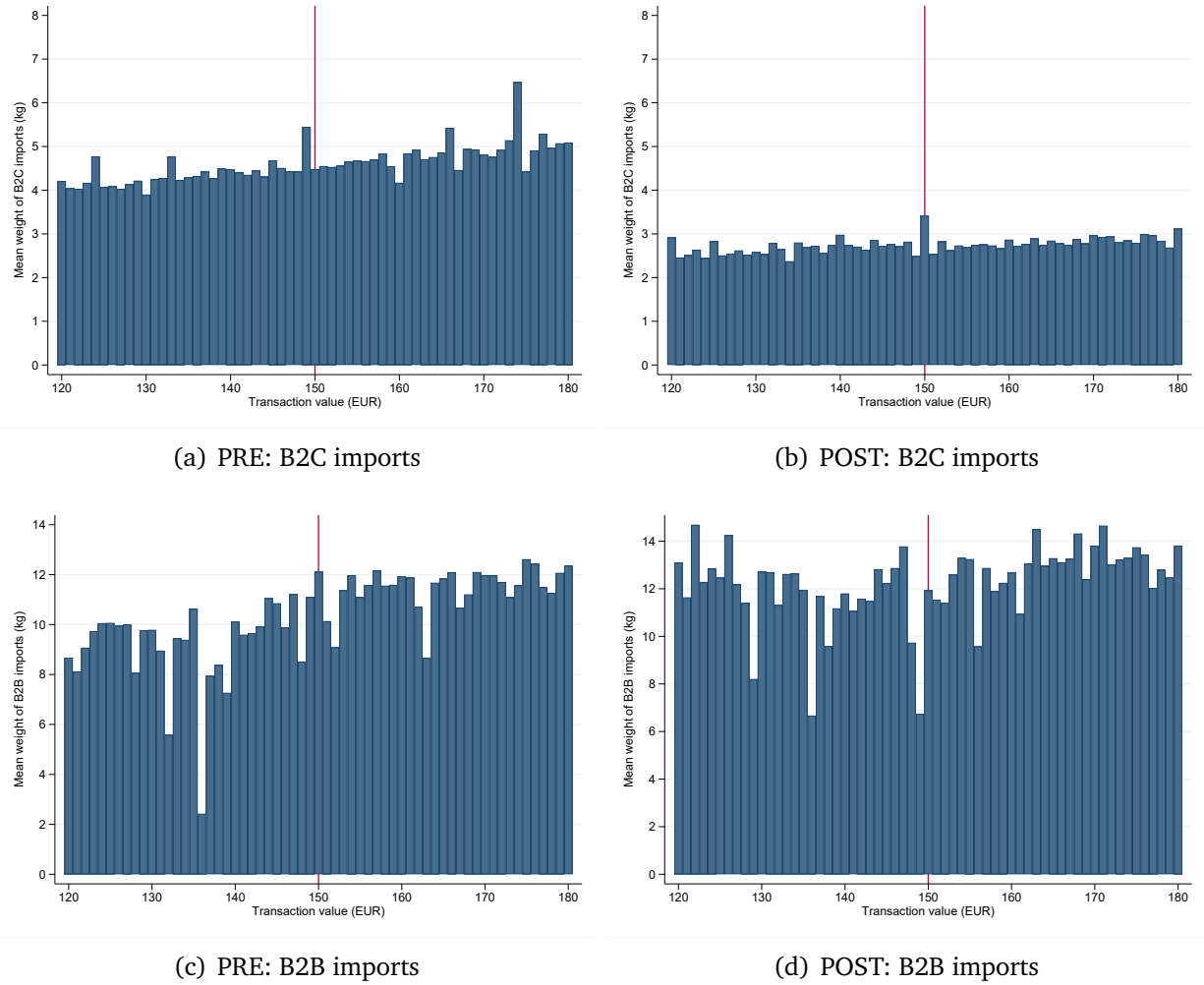
Notes: This figure displays histograms of transaction frequencies around the IOSS eligibility threshold for B2B imports from China and the United States, separately for the pre- and post-reform periods. Panels (a) and (b) show the distribution of B2B imports from China in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding distributions for B2B imports from the United States. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The vertical line marks the EUR 150 IOSS eligibility threshold. The corresponding B2C distributions for both countries are reported in Figure 7.

Figure B.5: IOSS – Distribution of B2C and B2B Transaction Values from China and USA in Percent



Notes: This figure displays histograms of relative transaction frequencies around the IOSS eligibility threshold for B2C and B2B imports from China and the United States for the pre- and post-reform periods. Panels (a) and (b), respectively. Panels (c) and (d) show the corresponding distributions for B2C and B2B imports from the United States. The distribution is restricted to transactions valued between EUR 120 and EUR 180, with each bin corresponding to a EUR 1 interval. The vertical line marks the EUR 150 IOSS eligibility threshold.

Figure B.6: IOSS – Distribution of Mean Weight of B2C and B2B Imports by Transaction Value



Notes: This figure plots mean transaction weights by declared transaction value for B2C and B2B imports, separately for the pre- and post-reform periods. Panels (a) and (b) show mean weights for B2C imports in the pre-reform period (2018 to 2020) and post-reform period (2022 to 2023), respectively. Panels (c) and (d) show the corresponding mean weights for B2B imports. The sample is restricted to transactions valued between EUR 120 and EUR 180, with each EUR 1 bin representing the average weight of all transactions declared at that value. Transactions above the 75th percentile of the pre-reform weight distribution are excluded separately for each group (25 kg for B2C, 95kg for B2B).